
Monte Carlo Counterparty Credit Risk Engine

Methodology, calibration and results for the ESF derivatives book

Prepared for Capitolis | Berkeley MFE Industry Project

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Abstract

This report describes a Monte Carlo counterparty credit risk (CCR) engine for Capitolis' equity swap financing (ESF) derivatives book, which comprises 16 trades with three counterparties. The engine simulates the joint evolution of the USD short rate (one-factor Hull-White model), 37 equities and USDJPY (correlated geometric Brownian motion), reprices every trade in every scenario with the supplied pricer library, and reports Expected Exposure (EE), median PFE, Potential Future Exposure at the 99th percentile (PFE99) and Maximum PFE (MPE) by counterparty and for the portfolio, prices counterparty credit risk (CVA) with bond-implied credit spreads, computes the Basel SA-CVA capital requirement, and provides the full set of sensitivities (Greeks) of the book and of the exposure measures. The report sets out the underlying concepts from first principles, the market data and calibration, each modelling choice together with the alternatives considered, the validation performed, and the results. Every quantity is either measured from the engine on real market data as of 2026-08-28 or is an explicitly stated assumption.

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Summary of results

The table reports the portfolio results for the uncollateralized book (3,000 Latin Hypercube scenarios, PFE at the 99th percentile).

Measure	Value	Meaning
Current exposure (portfolio)	\$107.7M	Loss if every counterparty defaulted today, after netting
Peak EE	\$121.9M at 2026-10-16	Highest average future exposure
Peak median PFE	\$121.3M at 2026-10-16	Typical (50th percentile) exposure at its highest date
Maximum PFE99 (MPE)	\$174.7M at 2026-10-09	Highest 99th-percentile exposure over the life of the book
Concentration	99% of current exposure is CPTY_C	A single \$500M bond forward (BF_0003) dominates
Time profile	Most exposure has run off by December 2026	Trades mature; a small Bond TRS tail runs to January 2028
Greeks	t = 0 book Greeks; sensitivities of EE, median PFE and PFE99 to equities, USDJPY, USD rates and volatilities	Complete, by netting set (Section 9)
CVA (BBB proxy)	\$210,394	Price of counterparty default risk, unilateral, uncollateralized (Section 8)
SA-CVA capital	\$1.93M (RWA \$24.1M)	Basel standardised approach; dominated by counterparty credit spread risk

Principal modelling choices (each is justified in Section 10):

- Monte Carlo under the risk-neutral measure. USD short rate: one-factor Hull-White (exact fit to today's SOFR curve). Equities and USDJPY: correlated geometric Brownian motion driven by the simulated rate.
- Correlation: one static 39x39 matrix estimated from 613 aligned daily returns and applied through a Cholesky factor; a PCA factor model (5 factors) is provided as an alternative.
- Sampling and size: Latin Hypercube sampling (lowest error of the five methods tested) with $N = 5,000$ scenarios for standard reporting (PFE99 relative standard error about 1% at the worst-case date), $N = 1,000$ for iteration and $N = 10,000$ for limit sign-off (Section 5.5).
- Dates: standard market pillar dates (O/N to 10Y) with every trade's own reset and maturity date forced onto the grid.
- Mean reversion: $a = 0.0167$ for USD from the swaption cube; for JPY the lower bound $a = 0.001$, because real JPY volatility rises with tenor (three independent sources, including the full daily OIS history).
- JPY: a negative-rate-capable Hull-White factor is built (sigma from real TONA; correlation with the USD rate calibrated at approximately zero) but does not yet drive JPY equity drift.
- The book is treated as uncollateralized (no CSA data); an MPOR-shifted collateralized calculation is built and demonstrated as a hypothetical.
- CVA: unilateral regulatory CVA (MAR50.32) with counterparty spreads proxied from ICE BofA bond indices by an assumed BBB rating and 60% LGD; SA-CVA capital from common-random-number bump sensitivities (Section 8).

1 The problem from scratch

1.1 What is counterparty credit risk?

A derivative is a contract whose value changes as markets move. If we hold a contract with positive value to us, the counterparty owes us that value. If the counterparty then defaults, we lose (some of) it. **Counterparty credit risk (CCR)** is that loss risk. It is not the same as market risk: market risk is about our profit and loss moving; CCR is about whether the other side can pay when the contract is in our favour.

Two properties make CCR different from ordinary risk measures:

- **It is one-sided.** If a contract is worth -5M to us we owe them; their default does not cost us that 5M back. So exposure is $\max(V, 0)$, a call option on the value (Figure 1).
- **It is about the future.** A default can happen at any date, so what matters is not today's value but the distribution of what the book could be worth on every future date. That needs simulation.

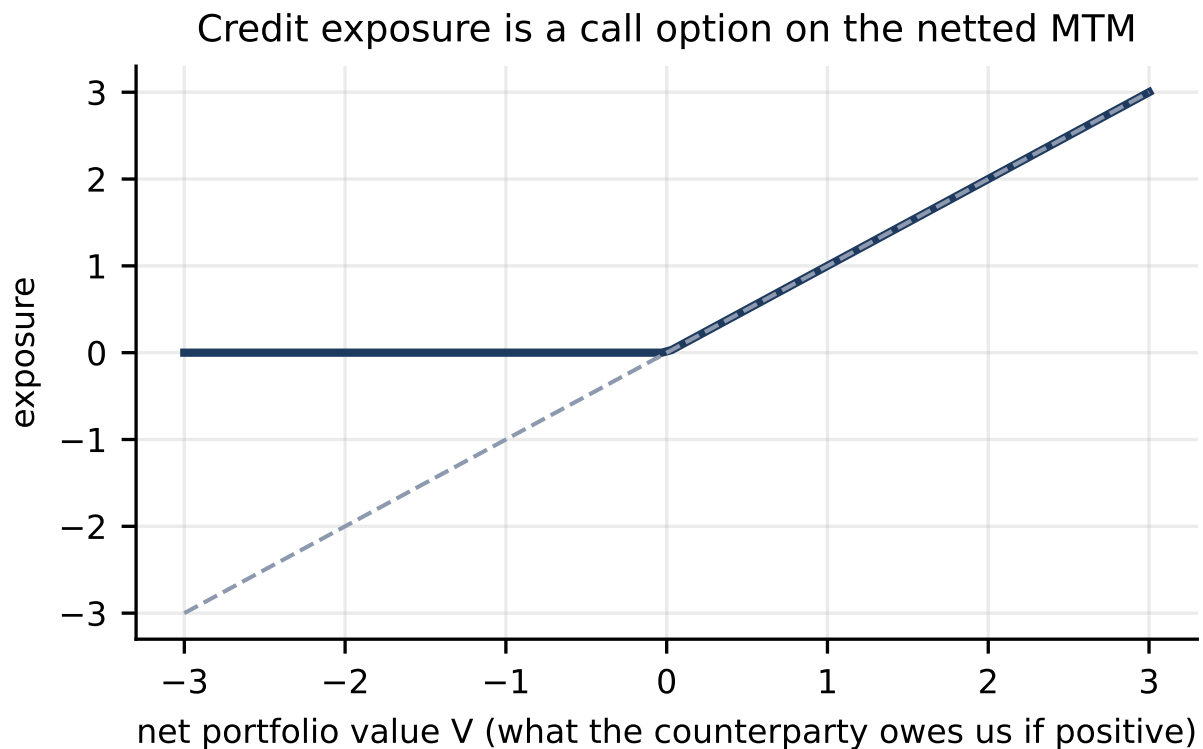


Figure 1: Exposure is the positive part of the net value V . Negative value is not a credit exposure.

1.2 Netting

Trades with the same counterparty under one legal netting agreement are combined before applying $\max(V, 0)$: a +10M trade and a -6M trade give exposure 4M, not 10M. Netting never crosses counterparties. The portfolio exposure is therefore the *sum over counterparties* of each counterparty's

own $\max(\text{sum of trade values}, 0)$.

1.3 The four measures reported

Measure	Definition	How to read it
Exposure profile $V_+(t)$	$\max(\text{net MTM}, 0)$ at each future date t , in each simulated scenario	The raw object; everything below summarises its distribution across scenarios
EE (Expected Exposure)	Mean of exposure across scenarios at date t	Average loss if default happens at t ; used for pricing credit charges (CVA)
Median PFE	50th percentile of exposure across scenarios at date t	The typical scenario. Exposure is floored at zero: the median is below EE when exposure is right-skewed (most scenarios small, a few large) and can be exactly zero when most scenarios are out of the money; it can sit slightly above EE when the book is almost always in the money, as for the dominant CPTY_C forward
PFE99 (Potential Future Exposure)	The 99th percentile of exposure at date t	In 99 of 100 scenarios exposure at t is below this level. Used to set limits
MPE (Maximum PFE)	Peak of the PFE curve over all dates	One number to set a counterparty limit against

Confidence level. PFE is reported at the 99th percentile throughout, together with the median PFE (the 50th percentile of the same exposure distribution), which describes the typical scenario.

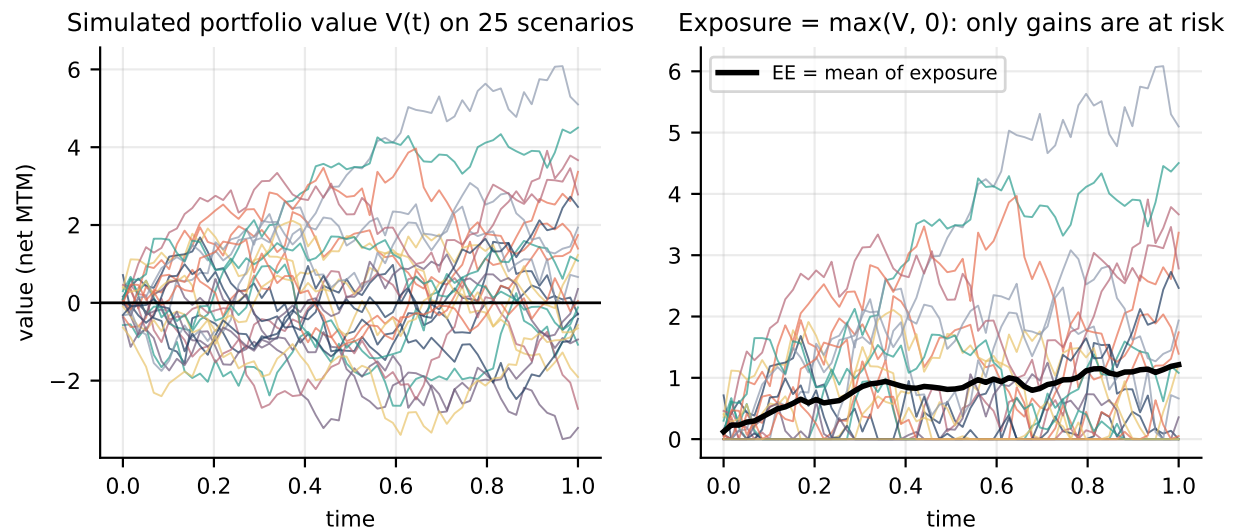


Figure 2: Left: simulated values of a portfolio through time. Right: the same paths after applying $\max(V, 0)$; EE is their average.

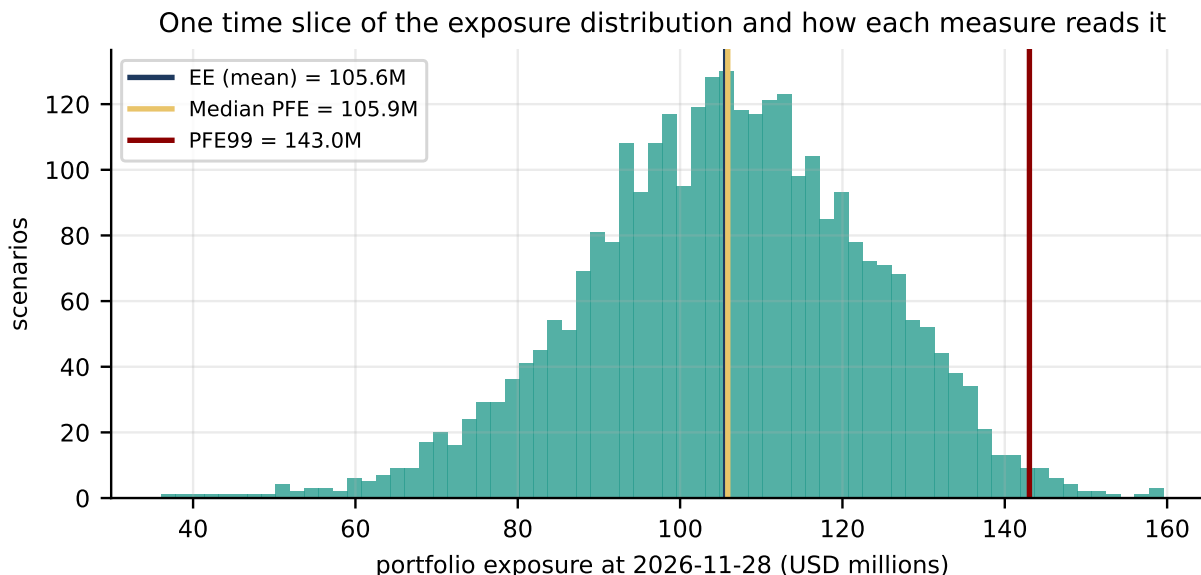


Figure 3: The real portfolio exposure distribution ~ 3 months out, with EE, median PFE and PFE99 marked. The long right tail is why EE, median PFE and PFE99 differ so much.

1.4 Why Monte Carlo

The value of the book at a future date depends on the future levels of ~ 40 market variables (37 stock prices, USDJPY, interest rates) that move together. There is no closed formula for the distribution of a netted, path-dependent-schedule book. Monte Carlo solves this by: (1) simulating thousands of joint futures for all the variables; (2) fully re-pricing every trade in every simulated future at every date; (3) netting; (4) reading off the distribution. The cost is that the answer has sampling noise, which we measure and control (Sections 5.4 and 5.5).

1.5 Real-world versus risk-neutral simulation

Simulating under the *risk-neutral* measure (drift = short rate minus dividends) is the pricing-consistent choice used for CVA and for exposure numbers that must be consistent with today's market prices. The alternative, real-world (physical) drift, is used for regulatory PFE limits and would add a risk premium to the drifts. We use risk-neutral because it needs no extra unobservable inputs and is verified by a martingale test (Section 11). Because the horizon is short (under two years) and most positions are financed at the same rates, the difference between the two drifts is small relative to volatility.

2 The trade book

The book has **16 trades** with **3 counterparties** (CPTY_A, CPTY_B, CPTY_C), referencing 37 unique equities (31 USD-quoted, 6 JPY-quoted) and 5 US Treasury bonds. Trade files are supplied by Capitolis in trade_data/.

2.1 The three instruments

Equity Total Return Swap (TRS). One party pays the total return of a basket of shares (price change plus dividends) and receives a funding leg (SOFR plus a spread, or a fixed rate) on a financed amount. Value = equity leg minus funding leg:

```
equity leg = sum_i shares_i * DF(end) * (F_i(end) - basis_i)    F(t) = S(t)/DF(end)
funding leg = sum_k rate_k * Notional * tau_k * DF(t_k)          rate_k = SOFR_fwd + spread (or fixed)
NPV = equity leg - funding leg
```

Eight of these (EQTRS_0001-0008), all ‘pay_equity’ from our side, so we are long the funding leg and short the equity return: our value rises when the shares fall. Two are JPY *compo* trades: both legs settle in USD but the equity leg pays the return of the USD value of a JPY share ($S_JPY \times USDJPY$), so equity and FX risk both matter.

Bond Forward. An agreement to sell/buy a Treasury at a fixed clean price on a future date: $NPV = \text{sign} \times (\text{notional}/\text{par}) \times DF(T) \times (\text{forward clean price} - \text{strike})$. The forward price is the spot financed at the repo rate less coupons in the window. Four of these, all short. BF_0003 has a 500M notional and dominates CPTY_C.

Bond TRS. Same idea as an equity TRS but on a bond: return leg (price change plus coupons) versus a funding leg over a reset schedule. Four of these.

Trade	Type	Counterparty	Ends	Today's NPV
EQTRS_0001	EquityTRS	CPTY_A	2026-10-14	-8,377,700
EQTRS_0002	EquityTRS	CPTY_A	2026-10-09	-2,079,657
EQTRS_0003	EquityTRS	CPTY_A	2026-10-16	11,151,337
EQTRS_0004	EquityTRS	CPTY_B	2026-11-02	-1,051,939
EQTRS_0005	EquityTRS	CPTY_B	2026-10-09	38,682
EQTRS_0006	EquityTRS	CPTY_B	2026-10-30	1,361,371
EQTRS_0007	EquityTRS	CPTY_C	2026-10-23	826,403
EQTRS_0008	EquityTRS	CPTY_C	2027-07-22	1,947,022
BF_0001	BondForwardTrade	CPTY_A	2026-10-30	512,531
BF_0002	BondForwardTrade	CPTY_A	2026-11-08	-432,501
BF_0003	BondForwardTrade	CPTY_C	2026-12-06	101,417,649
BF_0004	BondForwardTrade	CPTY_C	2026-10-25	1,973,388
BTRS_0001	BondTRS	CPTY_A	2028-01-15	123,516
BTRS_0002	BondTRS	CPTY_B	2027-01-15	221,019
BTRS_0003	BondTRS	CPTY_B	2027-04-09	28,948
BTRS_0004	BondTRS	CPTY_C	2026-11-04	37,741



Figure 4: Life span of every trade. Nearly all exposure will therefore disappear within four months; only BTRS_0001 and EQTRS_0008 run past mid-2027.

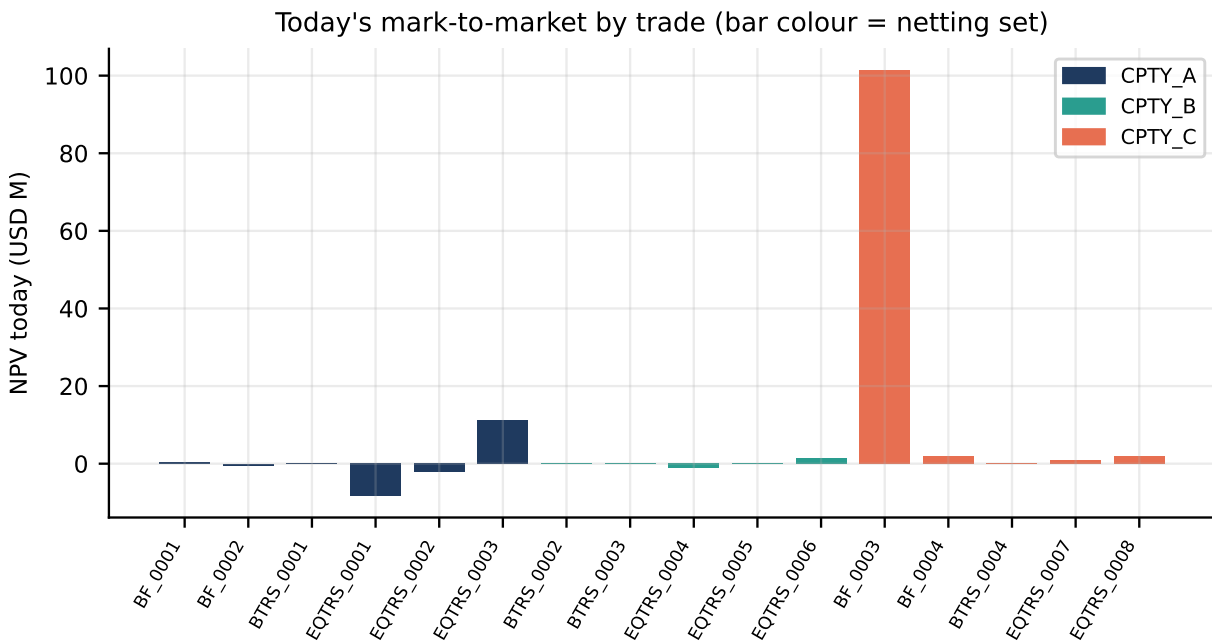


Figure 5: Today's mark-to-market per trade. CPTY_C's total is dominated by BF_0003 (+101M); CPTY_B nets to a small negative, so it has zero exposure today.

2.2 Pricing library (the ‘contract’)

The trades are priced by `capitolis_pricers`, a self-contained standard-library re-implementation of QuantLib/ORE-style discounted cash flows supplied with the project. Every pricer is a pure function `npv = pricer.npv(MarketState)`. Our simulation therefore never contains pricing logic of its own: at every (scenario, date) we build a `MarketState` from simulated inputs (a discount curve, equity spots, an FX curve) and call the same `.npv()`. This is a deliberate design choice: the pricers were independently reviewed (docs/notes/pricer_review.md), so any error in the exposure numbers cannot hide inside re-implemented pricing.

3 Market data: what we used and why

Every risk factor needs both a *starting value* and *dynamics* (volatility, correlation, mean reversion). We prefer real, verifiable data; where none exists we use a labelled assumption.

Input	Source	Used for	Note
USD discount curve	Databento: CME SR3/SR1 SOFR futures (33 live contracts), bootstrapped by us	Discounting, forwards, Hull-White fit	Validated against Treasury.gov par curve (spread 13bp at 1M to 46bp at 10Y)
37 equity spots and dividend yields	yfinance	GBM start values and drift	Keyed by ISIN; BRK.B ticker mismatch found and fixed
USDJPY spot	yfinance	FX start value	
Volatilities (39)	Realised, 3 years of daily data	Diffusion size	No options data available; documented proxy
Correlation matrix 39x39	613 dates where all series exist (inner join)	Cholesky / PCA	Verified positive semi-definite
SOFR level history	FRED	Rate vol, USD-JPY correlation	From April 2018
JPY OIS par-rate history, 35 tenors, 2011-2026	Bloomberg export provided by the project team (data/raw/sources/JPY.xlsx)	JPY curve, JPY factor volatility, JPY-USD differential, mean-reversion test	Licensed data: only derived quantities are reported
USD and JPY swaption cubes, USDJPY forwards	Bloomberg one-time export, 2026-08-31 snapshot	USD mean reversion, cross-checks, FX forwards	Licensed data: only derived quantities are reported
TONA (JPY overnight rate)	Bank of Japan public API, daily from 1998	JPY rate vol, USD-JPY correlation	Includes real negative-rate years
JGB par yields 1Y-40Y	Japan Ministry of Finance, daily from 1974	JPY mean-reversion test	Public

3.1 The USD curve

Databento provides the futures, not a ready OIS curve, so we build one: each 3-month SOFR future price gives an implied forward rate ($100 - \text{price}$) for its reference quarter; chaining consecutive quarters multiplies discount factors. Beyond the last live contract (about 6.3 years out) the curve is extrapolated flat. Two real bugs were found here by cross-checking with the Treasury curve (Section 12). The curve is the starting point of the Hull-White model and the discount curve at $t=0$.

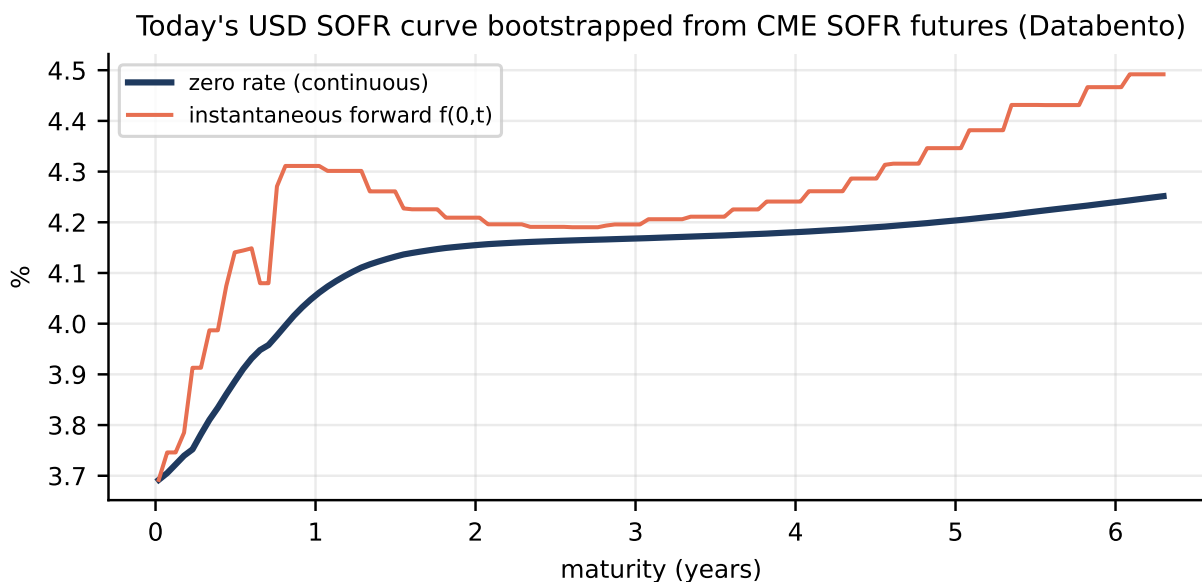


Figure 6: The bootstrapped USD SOFR curve. The upward slope (zero rate about 3.7% at the front, above 4% by 6 years) means forwards exceed spot rates.

3.2 Volatilities

Volatility is the annualised standard deviation of daily changes over the last three years (log returns for equities and FX; simple differences for the rate, because rate LEVELS near or below zero make log returns meaningless). Rate vol: 0.63% per year (normal vol on SOFR).

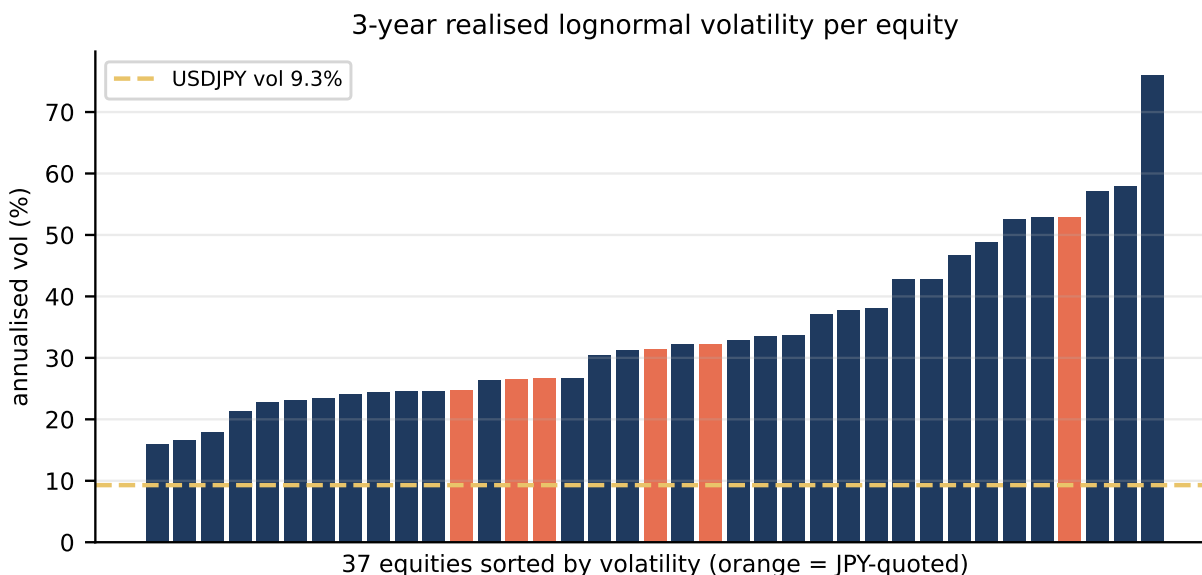


Figure 7: Realised volatility spans an order of magnitude across the 37 names. A few names carry 50-76% vol, which dominates tail exposure of the trades that reference them.

Choice and alternative. Implied volatility from options would be forward-looking and is the norm

for pricing, but we had no single-name options data (the Bloomberg export has only SPX/TOPIX index vols, which would need a per-name basis assumption). A 3-year realised window is a documented, reproducible proxy. Its limitation is that it is backward-looking and cannot see regime changes; the sensitivity test (Section 11) shows how much the results move if vols are 50% higher.

3.3 Correlations

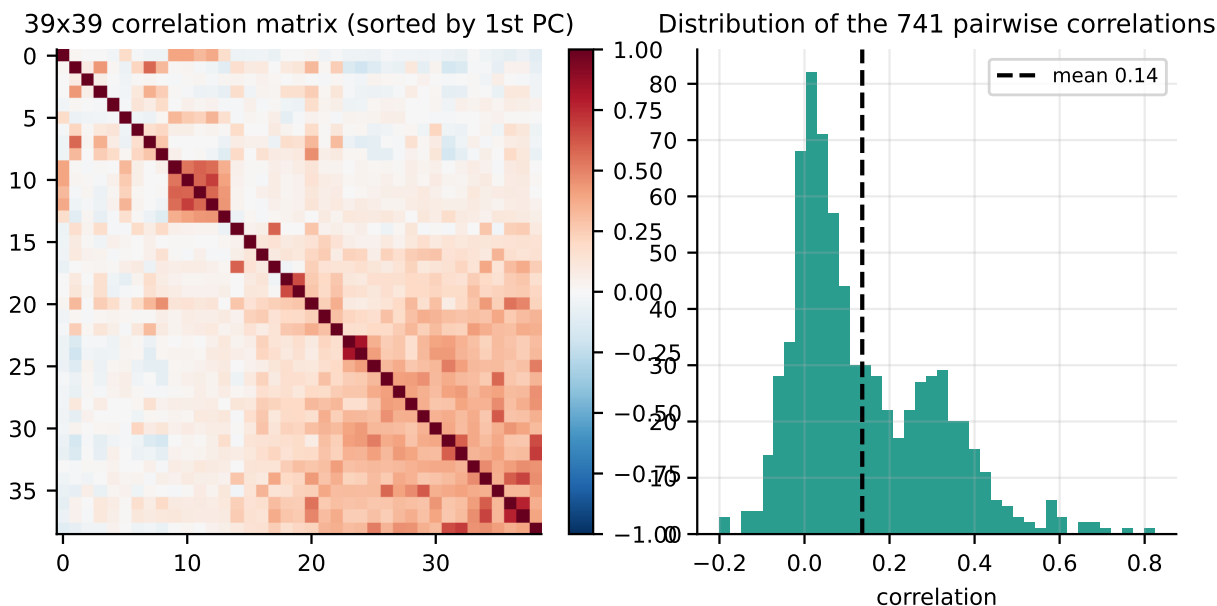


Figure 8: Left: the 39x39 correlation matrix reordered so similar names sit together; a broad positive ‘market’ block is visible. Right: the 741 pairwise correlations centre on 0.14.

Estimation and stationarity. The matrix is static, not re-estimated per date. We take the daily returns of all 39 factors, keep only the 613 dates on which every series has a price (US and Tokyo trade on different calendars, so series are joined by calendar date rather than timestamp), and compute one pairwise correlation matrix. It is assumed constant over the simulation horizon. The alternatives (time-varying DCC-GARCH, or stressed correlations) add parameters we cannot validate with our data; we flag this as a limitation.

4 The risk-factor models

4.1 USD short rate: Hull-White one-factor

We need the whole USD curve at every future date and scenario, because bonds and funding legs are discounted with it. The standard tool is the **Hull-White one-factor model**, a Gaussian, mean-reverting short-rate model:

$$dr(t) = [\theta(t) - a \cdot r(t)] dt + \sigma dW(t)$$

a is mean reversion (how fast rates are pulled back), σ is rate volatility, and $\theta(t)$ is chosen so that the model prices today's curve exactly. We use the equivalent shifted Ornstein-Uhlenbeck form $r(t) = x(t) + \alpha(t)$:

$$\begin{aligned} x(t+dt) &= x(t) e^{-a dt} + \sigma \sqrt{(1 - e^{-2a dt}) / (2a)} Z && \text{(exact transition)} \\ \alpha(t) &= f(0,t) + \sigma^2 / (2a^2) (1 - e^{-a t})^2 && \text{(fits today's curve)} \\ P(t,T) &= A(t,T) \exp(-B(t,T) r(t)), \quad B = (1 - e^{-a(T-t)})/a && \text{(analytic bond price)} \end{aligned}$$

Why this model. (1) It reproduces today's curve to 1e-9 (tested). (2) Bond prices at future nodes are analytic, so building the discount curve for a scenario costs one function evaluation (10x faster than building a curve object, and exact rather than interpolated). (3) It is Gaussian, so it can produce negative rates, essential for JPY history and harmless for USD. **Alternatives rejected:** CIR/Black-Karasinski (positive-only, cannot represent negative JPY rates); LMM/HJM multi-factor curve models (more realistic curve twists but many more parameters than our data can support and slower); a constant-rate assumption (would ignore the funding-leg and bond-forward rate risk).

Parameters. $\sigma = 0.63\%$ (realised SOFR vol); $a = 0.0167$ (calibrated, Section 6). Today's short rate $r(0) = 3.69\%$.

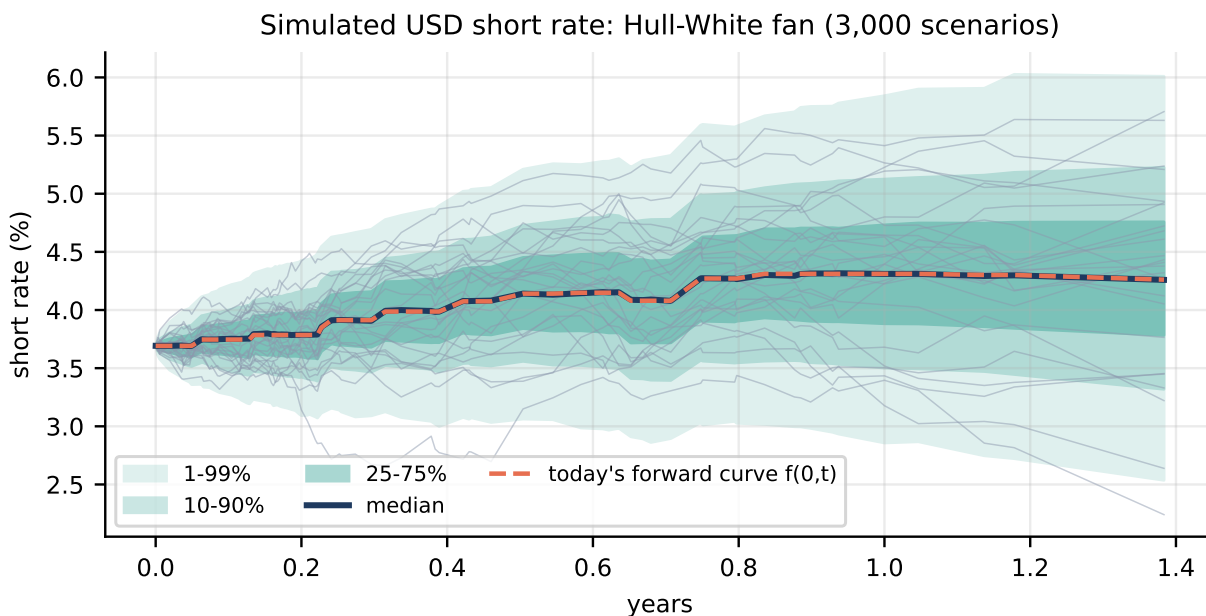


Figure 9: Simulated USD short rate. The median tracks today's forward curve (dashed) because the model is fitted to it; the widening bands are rate uncertainty.

4.2 Equities and USDJPY: correlated geometric Brownian motion

$dS_i / S_i = (r(t) - q_i) dt + \sigma_i dW_i$	USD-quoted names
$dS_i / S_i = (r_{JPY}(t) - q_i) dt + \sigma_i dW_i$	JPY-quoted names
$dFX / FX = (r_{USD} - r_{JPY}) dt + \sigma_{FX} dW_{FX}$	USDJPY (JPY per USD)

Each step is the exact log-normal step $\ln S(t+dt) = \ln S(t) + (r - q - \sigma^2/2) dt + \sigma \sqrt{dt} Z$, with r the simulated USD short rate at the start of the step (so equities and rates are linked). Under the risk-neutral measure the expected growth is $r - q$, verified in Section 11 by a martingale test.

Why GBM. It matches the lognormal vol convention we measure, is the market default for equity CCR, and needs only a vol per name. **Alternatives:** local/stochastic vol (Heston, SABR) would capture skew and vol-of-vol but need option surfaces we do not have; jump models help tails but add unobservable parameters. We disclose that GBM understates fat tails.

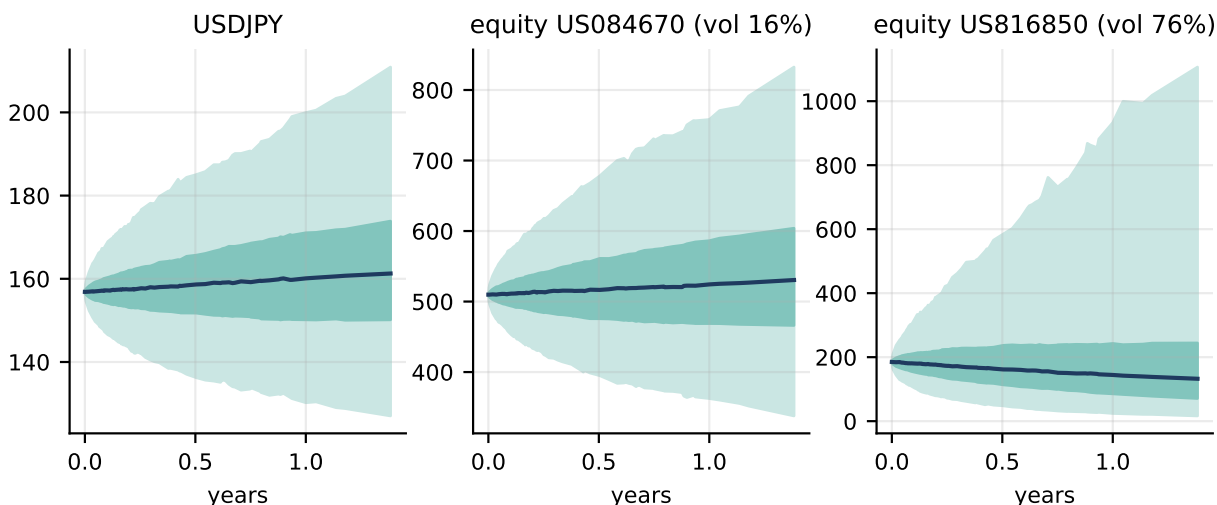


Figure 10: Simulated USDJPY and two equities (a low-vol and the highest-vol name) with 1-99 and 25-75 percent bands.

4.3 Correlation and the Cholesky factor

Independent random numbers give independent assets. Real assets move together, so we transform independent standard normals Z into correlated ones. Given the target correlation matrix C , the **Cholesky decomposition** finds the lower-triangular L with $L L' = C$. Then $Y = L Z$ has exactly correlation C :

$$\text{Cov}(Y) = L \text{Cov}(Z) L' = L I L' = L L' = C$$

It is used because it is the cheapest exact way to impose an arbitrary positive-definite correlation structure. A tiny eigenvalue clip guards against CSV round-off making C numerically non-PSD.

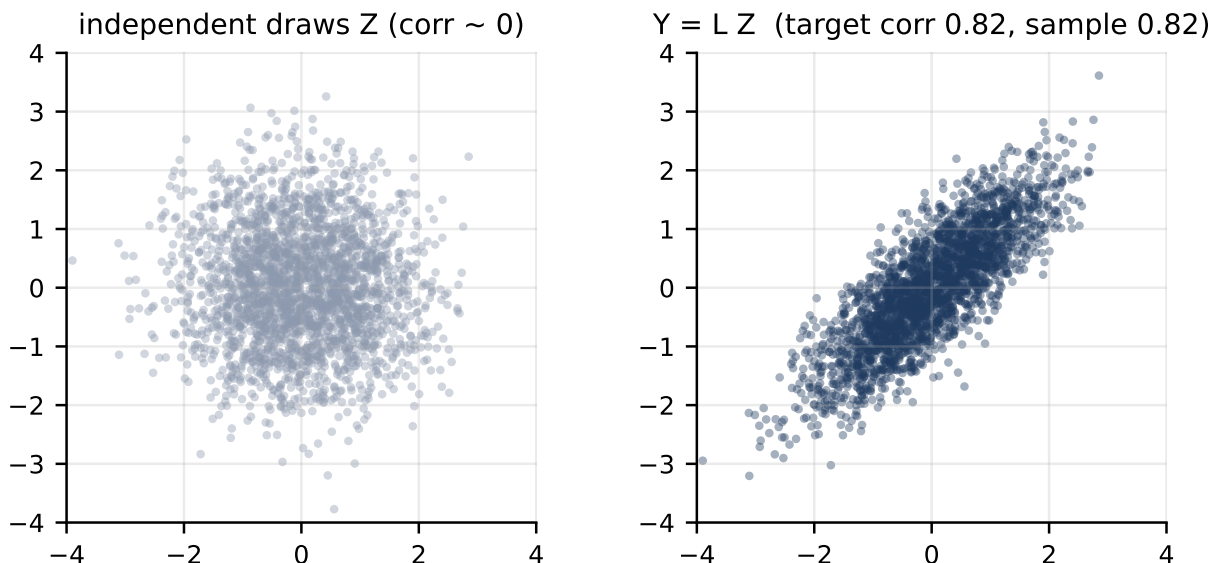


Figure 11: Demonstration on the most correlated pair in our real matrix (US21037T / US92840M, corr 0.82): independent draws (left) become correlated draws (right) after multiplying by L .

4.4 Optional alternative: a PCA factor model for correlation

A 39x39 matrix has 741 free correlations estimated from only 613 days, so many entries are noisy. A **factor model** assumes co-movement comes from a few common drivers plus each name's own noise: $C \sim B B' + \text{diag}(d)$, where B ($39 \times k$) are loadings on k factors. We build it by principal-component analysis: $B = \text{top-}k \text{ eigenvectors} \times \sqrt{\text{eigenvalue}}$, and $d = 1 - \text{row sums of squares}$ so every name still has unit variance. Simulation then draws k systematic and 39 idiosyncratic shocks.

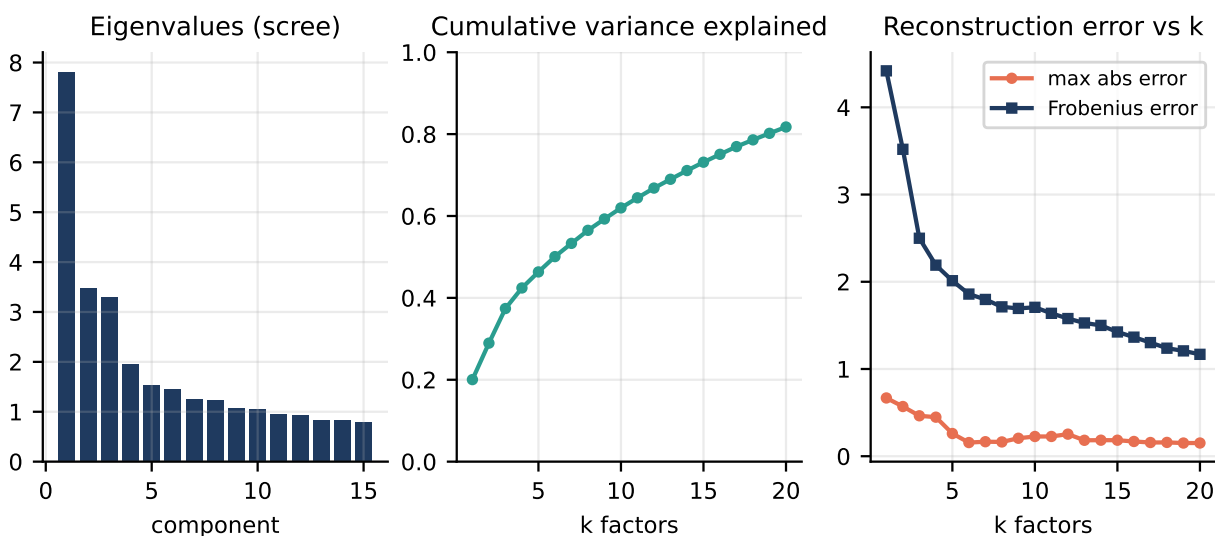


Figure 12: Left: the first eigenvalue (7.8 of a total 39) is one dominant market factor. Middle: 5 factors explain 46% of variance and 10 explain 62%. Right: reconstruction error falls as k grows (exactly zero at full rank).

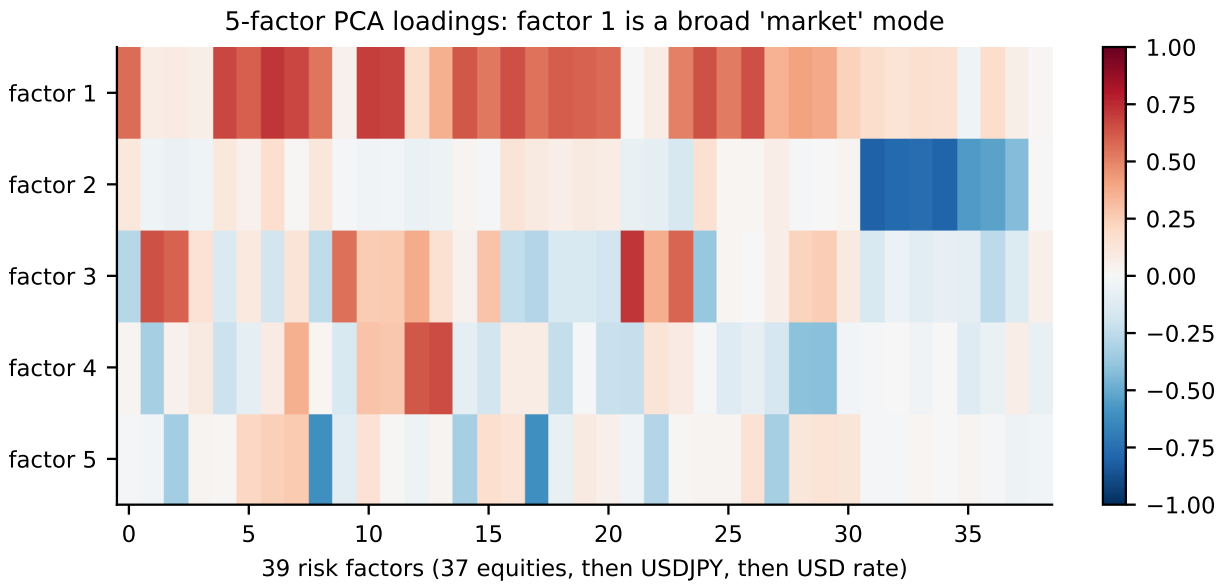


Figure 13: Loadings of each of the 39 factors on the first five principal factors.

Trade-off. The factor model is smoother and more robust to estimation noise and gives quasi-random sequences a low-dimensional space where they work best, but it only approximates the empirical matrix. Because it changes results, it is an option (`corr_mode='factor'`), not the default. Section 7.5 compares its exposure profiles with the full-rank engine.

4.5 JPY: a negative-rate-capable rate model

Until recently JPY drift used r_USD minus a constant differential. That cannot represent JPY's own rate dynamics. We therefore built a genuine second Hull-White factor for JPY (`build_jpy_hull_white`): it starts from the real JPY OIS curve and, being Gaussian, has **no floor at zero**.

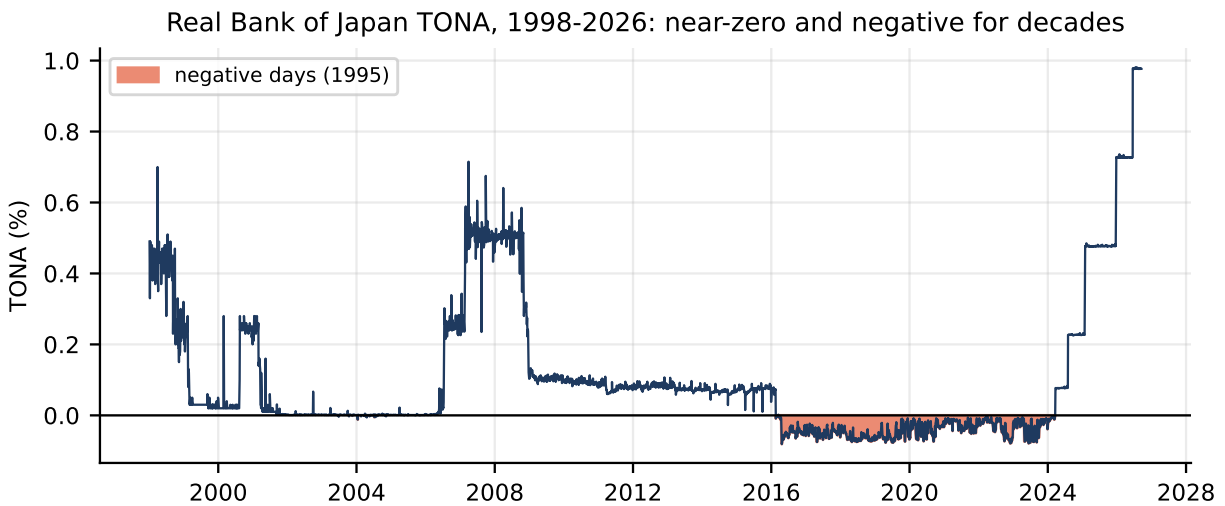


Figure 14: Real TONA history from the Bank of Japan: rates were at or below zero for most of 25 years, with 1995 negative daily fixings (2003 and 2016-2024). A model that floors at zero could not represent this.

The daily JPY OIS history provided by the project team (35 tenors, 3903 business days from October 2011) shows the same for the whole short end of the curve: the overnight call rate was negative on 2109 days, and OIS par rates out to five years were negative on 2340 days, with a minimum of -0.37%.

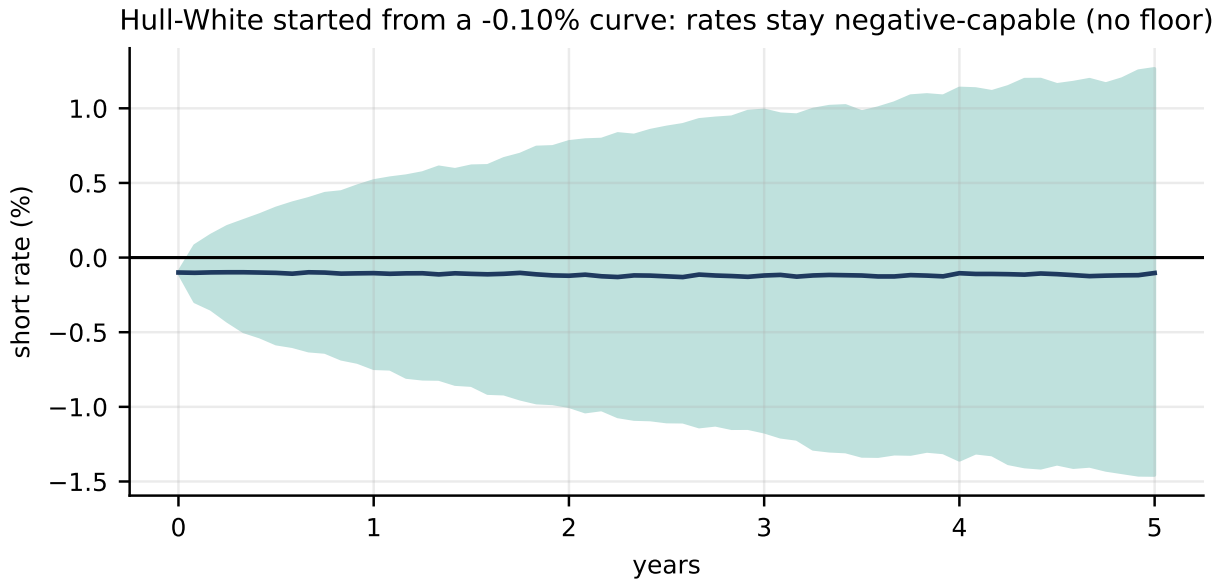


Figure 15: Demonstration of the capability: Hull-White started from a synthetic -0.10% flat curve. 63% of simulated points are negative and nothing is clipped. (Our only real JPY curve snapshot, Aug 2026, is positive after BOJ hikes, so the negative case is shown synthetically.)

JPY factor parameter	Value	Source / status
Curve	JPY OIS zero curve, 2026-08-28	Bootstrapped from the daily JPY OIS par history (project-team Bloomberg file); matches Bloomberg's own zero curve to within 1bp at all tenors
sigma	0.267%/yr	Realised vol of the overnight call rate, 3y window, from the same OIS file (Bank of Japan TONA gives 0.276%, consistent)
a (mean reversion)	0.0010	Lower bound: all three real JPY calibrations gave a negative a (Section 6.2)
USD-JPY rate-factor correlation	-0.040	Calibrated from real SOFR vs TONA daily changes (n=1983, p=0.07): statistically zero
Wired into simulate_paths()?	NO (disclosed)	JPY equity/FX drift still uses r_USD minus the constant differential (2.59%, our USD curve minus the JPY OIS zero curve at one year)

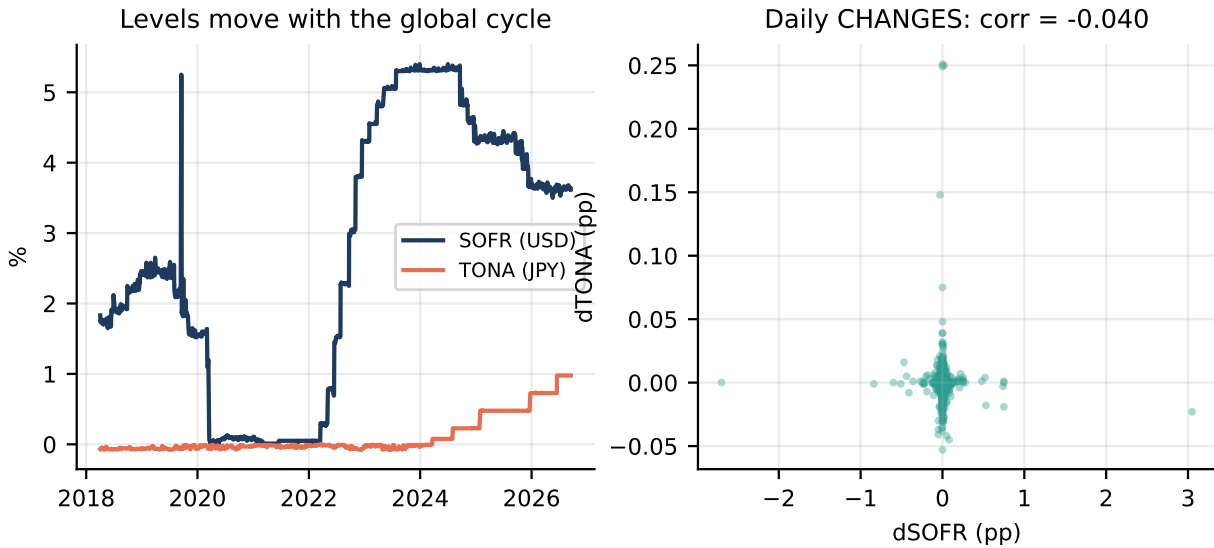


Figure 16: Levels (left) rise together with the global cycle (level correlation ~ 0.38), but day-to-day changes (right) are uncorrelated (-0.040): central banks set policy independently. A Hull-White correlation needs the CHANGES.

Why this matters little for the results: only 2 of 16 trades are JPY compo, so this simplification affects a small part of the book. It is disclosed rather than hidden, and the next engineering step is to let the simulated JPY rate drive those two trades.

5 How the simulation engine works

5.1 The pipeline

1. calibrate: curve, sigma, a, vols, correlation (real data)
2. grid: choose simulation dates (pillars + trade event dates [+ MPOR look-ahead])
3. draw: random numbers (Latin Hypercube) -> correlated shocks (Cholesky or PCA)
4. step: advance short rate, 37 equities, USDJPY along every path
5. reprice: for every (scenario, date): build MarketState, call npv() of all 16 trades
6. aggregate: net by counterparty -> max(.,0) -> EE, median PFE, PFE99, MPE

5.2 Choosing the simulation dates

Monthly steps are simple but wasteful and blunt: they spend nodes evenly when the risk changes fastest near term, and a trade's own reset or maturity rarely falls on a month node. Following Capitolis' guidance, we adopt **pillar dates**: the standard market curve tenors also used to build the SOFR curve (O/N, T/N, 1W, 2W, 1M, 2M, 3M, 6M, 9M, 1Y, 18M, 2Y ...). This mirrors production CCR practice: dense short-term, sparse long-term. In addition, every trade's own reset/settlement/maturity date is **forced onto the grid** so exposure jumps at cash-flow dates are not smoothed over.

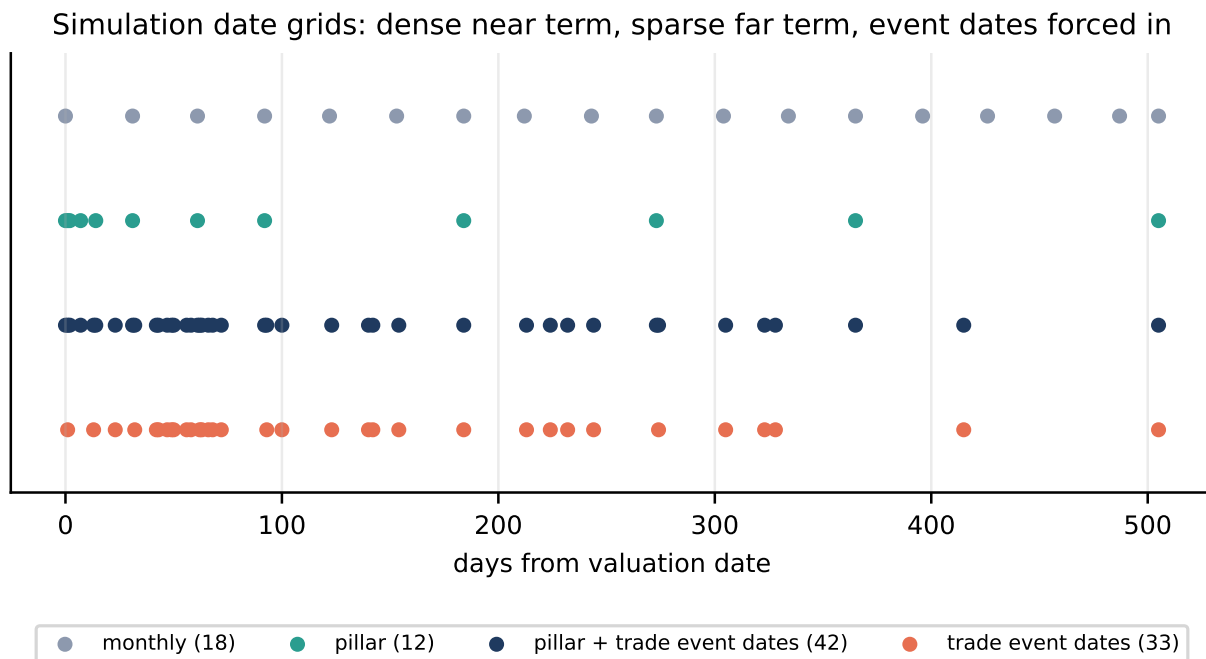


Figure 17: For this book the pillar grid (12 dates) plus forced trade-event dates (42 in total) is more compact than a plain monthly grid (18) and hits every real cash-flow date exactly.

5.3 Speed

Bottleneck	Fix	Before	After
Discount curve per (scenario, node)	Analytic Hull-White bond price instead of building a Curve object	0.206 ms/call	0.019 ms/call (10.6x, and exact)
Pure-Python pricers (~1.2 ms each)	Multiprocessing over scenarios, 8 workers (a Windows spawn re-import bug was fixed)	113 ms/scenario	51 ms/scenario at the time

Bottleneck	Fix	Before	After
Combined, 2,000 scenarios x 16 trades		~12 min projected	93 s (7.7x)

Current measured cost for the 3,000-scenario reporting run: 1 s simulation of paths, 682 s repricing (227 ms/scenario, including the extra MPOR look-ahead nodes).

5.4 Random numbers and variance reduction

Plain pseudo-random draws waste information: by chance a sample can cluster. Five sampling schemes sit behind one interface: **pseudo-random** (baseline), **antithetic** (each draw z paired with $-z$), **moment-matched** (rescale so the sample mean is 0 and std 1), **Sobol** (low-discrepancy quasi-random), and **Latin Hypercube** (each dimension split into equal-probability bins, one draw per bin).

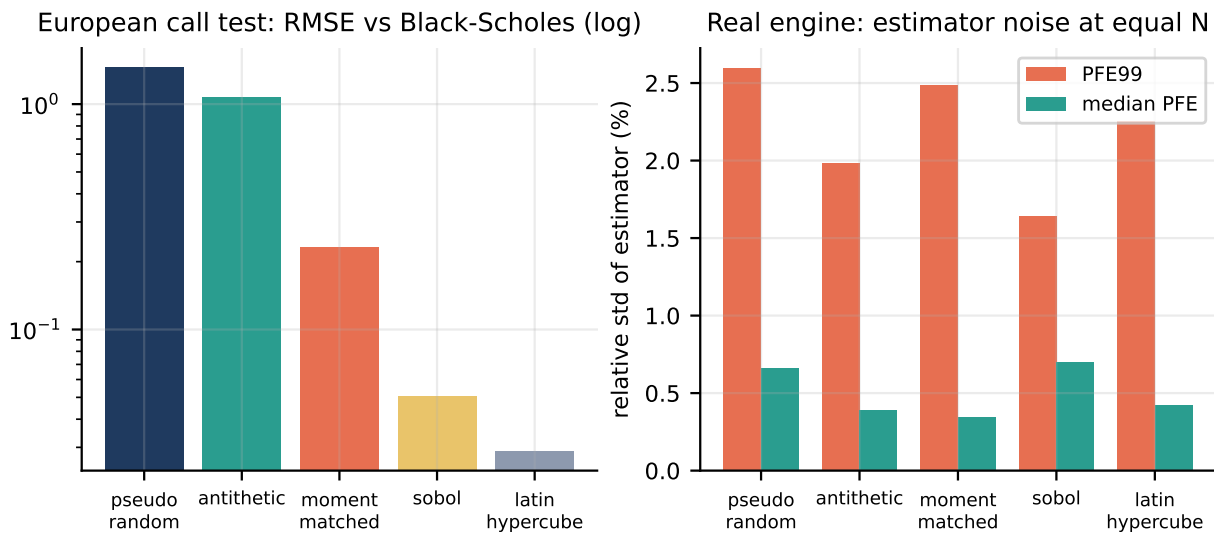


Figure 18: Left: on a European call with a known Black-Scholes value, Latin Hypercube has about 51x lower error than pseudo-random. Right: on the real 663-dimensional engine (300 scenarios, 8 repeats) the methods are much closer, and no method is best for both PFE99 and the median PFE.

Method	Call RMSE	vs pseudo-random	PFE99 rel. std (real engine)	Median PFE rel. std (real engine)	Verdict
pseudo random	1.4563	-	2.60% (1.00x)	0.66% (1.00x)	Baseline
antithetic	1.0786	1.4x better	1.98% (1.31x)	0.39% (1.69x)	Improves both measures modestly on the real engine, but only 1.4x on the controlled test
moment matched	0.2317	6.3x better	2.49% (1.04x)	0.35% (1.91x)	Helps the median only; no PFE99 benefit
sobol	0.0507	28.7x better	1.64% (1.58x)	0.70% (0.95x)	Best PFE99 on the real engine but no median benefit; unscrambled Sobol degrades in high dimension and is harder to replicate

Method	Call RMSE	vs pseudo-random	PFE99 rel. std (real engine)	Median PFE rel. std (real engine)	Verdict
latin hypercube	0.0288	50.6x better	2.25% (1.15x)	0.42% (1.57x)	SELECTED: 50x on the controlled test and improves both PFE99 and median PFE on the real engine

Relative standard deviation of the estimator across 8 independent repeats at 300 scenarios (portfolio, about 6 weeks out, near the PFE99 peak); the multiple in brackets is the improvement over pseudo-random. With only 8 repeats each standard deviation is itself uncertain by roughly a quarter, so real-engine differences between the better methods are indicative rather than conclusive.

Finding. The controlled test gives a clear ordering (Latin Hypercube, Sobol, moment matching, antithetic, pseudo-random). On the real engine, at 663 effective dimensions (39 factors x 17 time steps), the advantages compress and the ranking depends on the statistic: the tail (PFE99) and the centre (median PFE) respond to different methods. Latin Hypercube is the only method that is both best on the controlled test and better than pseudo-random on both real-engine statistics.

Decision: Latin Hypercube sampling. It is the best method on the controlled test and the most consistent across both reported statistics on the real engine, at negligible additional cost over plain pseudo-random draws. Sobol is the runner-up on the controlled test but is not recommended here because its benefit disappears for the median PFE and it is less robust at this dimensionality.

5.5 Number of scenarios

Monte Carlo error falls as $1/\sqrt{N}$. Rather than rerun the expensive repricing at every candidate N , we estimate the standard error by bootstrap resampling of a simulated pool. Two studies are combined: (i) resampling of the 3,000-scenario reporting run, at the date of peak PFE99, for both PFE99 and the median PFE; (ii) a dedicated 30,000-scenario pool for the 99th percentile at the one-year node, which extends the range to large N .

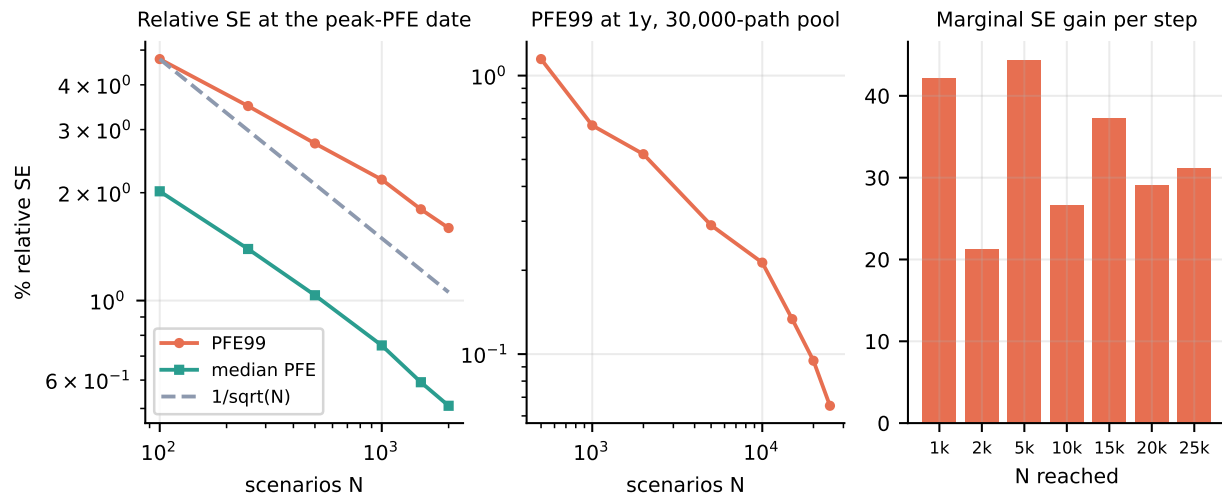


Figure 19: Left: relative standard error of PFE99 and median PFE at the peak-PFE date (2026-10-09); the error follows the $1/\sqrt{N}$ law. Middle and right: PFE99 tail study at one year; error is larger for the same N , and the marginal gain per additional batch peaks at $N = 5,000$ and then declines.

N	Relative SE of PFE99	Relative SE of median PFE
100	4.72%	2.02%
250	3.49%	1.39%
500	2.74%	1.03%
1,000	2.18%	0.75%
1,500	1.80%	0.59%
2,000	1.59%	0.51%

Relative standard errors at the peak-PFE date, bootstrapped from the reporting run (finite-pool corrected). Extrapolating with the $1/\sqrt{N}$ law, the 3,000-scenario run used for this report has a PFE99 relative standard error of about 1.24% at its peak.

N	PFE99 at 1y (USD)	Rel. SE at 1y	Rel. SE at peak date (est.)	Marginal SE gain (1y)	Est. time	Verdict
500	149,036	1.15%	3.03%	-	74s	Screening only
1,000	149,353	0.66%	2.14%	+42.1%	119s	Iteration and what-if runs
2,000	149,341	0.52%	1.52%	+21.2%	209s	Acceptable for routine monitoring
5,000	149,457	0.29%	0.96%	+44.4%	479s	RECOMMENDED: standard reporting
10,000	149,411	0.21%	0.68%	+26.6%	929s	Recommended for limit sign-off
15,000	149,477	0.13%	0.55%	+37.3%	1,379s	Diminishing returns
20,000	149,535	0.09%	0.48%	+29.1%	1,829s	Diminishing returns

N	PFE99 at 1y (USD)	Rel. SE at 1y	Rel. SE at peak date (est.)	Marginal SE gain (1y)	Est. time	Verdict
25,000	149,505	0.07%	0.43%	+31.1%	2,279s	Diminishing returns

Tail study at the one-year node (pool value \$149,482, 30,000 scenarios) and the corresponding estimate at the date of peak PFE99, where the exposure distribution is wider (relative SE = 2.1% / sqrt(N/1000), fitted to the reporting-run bootstrap). Times assume 8 cores.

Decision: number of paths. Use **N = 5,000** for standard PFE99 reporting: relative standard error 1.0% at the worst-case date (0.29% at the one-year node), about 8 minutes on 8 cores. Use **N = 1,000** for iteration and what-if runs (2.1% at the worst-case date, about 2 minutes) and **N = 10,000** when a limit is being signed off (0.7%, about 15 minutes). We do not recommend more than 15,000: cost grows linearly while error falls only as 1/sqrt(N), and the remaining sampling error (below 0.6%) is an order of magnitude smaller than the model sensitivities in Section 11.4 (a 50% increase in equity volatility moves MPE by about 9%). The figures in this report use N = 3,000 (1.2% at the worst-case date).

6 Calibrating Hull-White mean reversion

Mean reversion a controls how fast rate shocks decay; the convergence study flagged it as the largest unquantified model uncertainty. The textbook method is to fit swaption volatilities. Hull-White predicts that the volatility of a forward rate at maturity T decays with time to maturity:

$$\sigma_f(t, T) = \sigma \cdot \exp(-a(T - t)) \quad \Rightarrow \quad \ln(\text{vol}) = \ln(\sigma) - a \cdot \text{tenor} \quad (\text{line, slope} = -a)$$

So measuring vol at several tenors and regressing $\ln(\text{vol})$ on tenor gives a . We tried three real data routes, in order of preference:

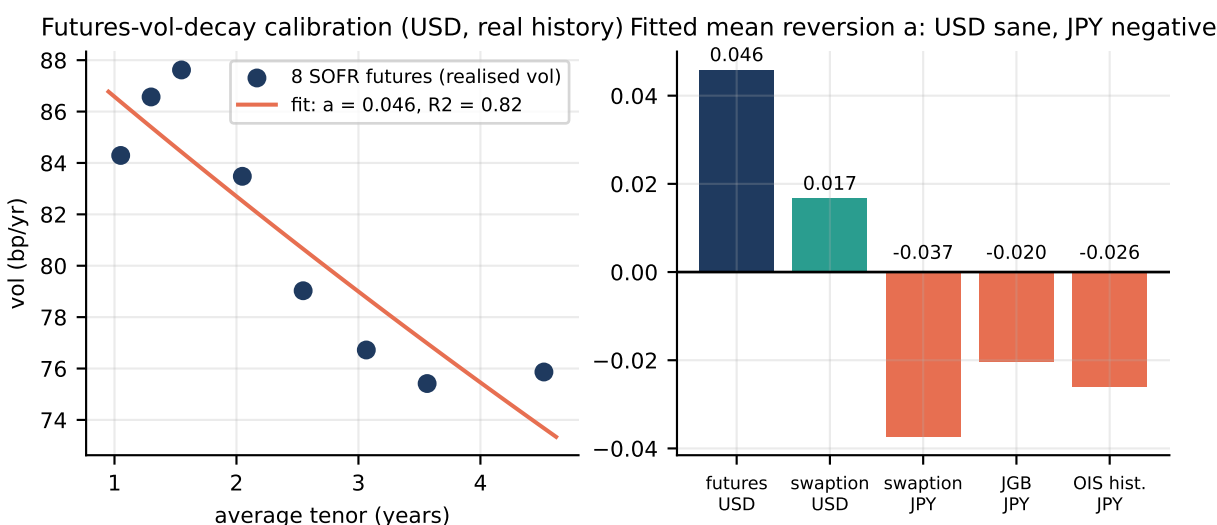


Figure 20: Left: USD SOFR-futures fit. Right: fitted a for each route. USD routes give sensible positive values; all three JPY routes give a NEGATIVE a .

Route	Data	a	R2	Verdict
Swaption cube, USD (preferred)	Bloomberg ATM normal vol, 1M expiry, tenors 1Y-15Y	0.0167	0.84	USED for USD
SOFR-futures vol decay, USD	8 contracts, ~2y Databento history	0.0458	0.82	cross-check (different instrument, higher a)
Swaption cube, JPY	Bloomberg JPY OIS ATM vols (one snapshot)	-0.0374	0.84	invalid (vol rises with tenor)
JGB yield vol, JPY	MOF daily 1Y-30Y yields, 3y window	-0.0203	0.49	invalid (vol rises with tenor)
JPY OIS history, JPY (most complete)	Daily OIS par rates, 35 tenors, 2011-2026, 1Y-30Y vols, 3y window	-0.0261	0.62	invalid (vol rises with tenor); lower bound $a = 0.001$ used

6.1 Why the two USD numbers differ

0.0167 (swaptions) versus 0.0458 (futures) is not a contradiction: they use different instruments (swaption-implied vs realised vol) and different tenor ranges (1-15Y swap tenors vs 1-3Y forward reset times). The swaption route is the industry standard and is used; the gap is itself a measure of model uncertainty and motivates the sensitivity test.

6.2 Why JPY does not fit, and what we did

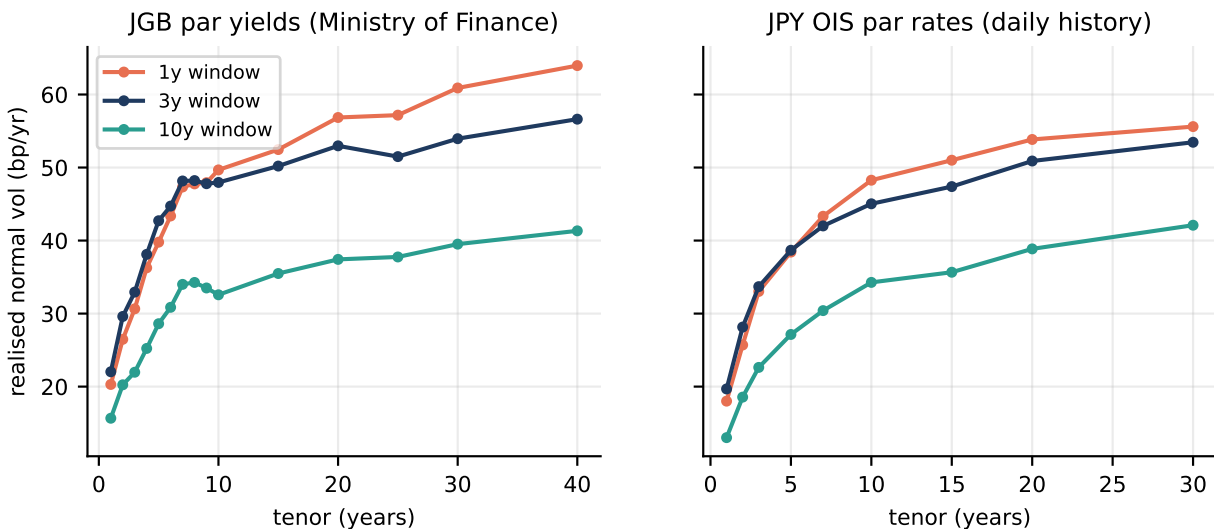


Figure 21: Realised volatility of JPY rates by tenor from two independent real datasets. At every window tried (1y, 3y, 10y) long tenors are MORE volatile than short ones, the reverse of the decay Hull-White assumes.

This is not bad data; it is a structural finding confirmed by three independent real sources: the daily JPY OIS curve history (35 tenors, 2011 to 2026, provided by the project team and the most complete of the three), 52 years of JGB yields, and a 2026 swaption cube. The fitted α is negative for the OIS history in every window tested (1, 3, 5, 10 and 15 years, and since the end of negative rates in March 2024). One factor, the level, explains about 84% of daily changes in the curve, and long rates move more than short rates. The plausible reason is that for two decades the Bank of Japan pinned the SHORT end (zero and negative policy rates, yield-curve control), so short-tenor volatility was suppressed while longer tenors moved more freely. A single mean-reverting Gaussian factor implies volatility that DEcreases with tenor, so it cannot represent this shape, and a negative α would make the short rate diverge.

Choice: use the lowest admissible mean reversion, $\alpha = 0.0010$, the Ho-Lee limit in which volatility is flat across tenors. This is the closest a Gaussian factor can get to the data, and it replaces our earlier use of the USD value. A two-factor or regime-dependent model would be needed to match the shape, and is listed under future work.

Practical impact: small. JPY affects two trades, the JPY factor is not yet wired into the drift, and the USD parameters are the ones used for discounting the whole book.

7 Results

7.1 Exposure today ($t = 0$)

Counterparty	Trades	Net MTM today	Netted exposure today	Comment
CPTY_A	6	897,526	897,526	Small positive net
CPTY_B	5	598,081	598,081	We owe them: zero exposure
CPTY_C	5	106,202,202	106,202,202	One 500M bond forward dominates
Portfolio	16	107,697,810	107,697,810	Sum of counterparty exposures (netting does not cross counterparties)

The simulated EE at $t=0$ equals the directly computed current exposure to 0.0000% (the engine's built-in self-check): at $t=0$ there is no randomness so both must agree exactly.

7.2 Exposure profiles through time

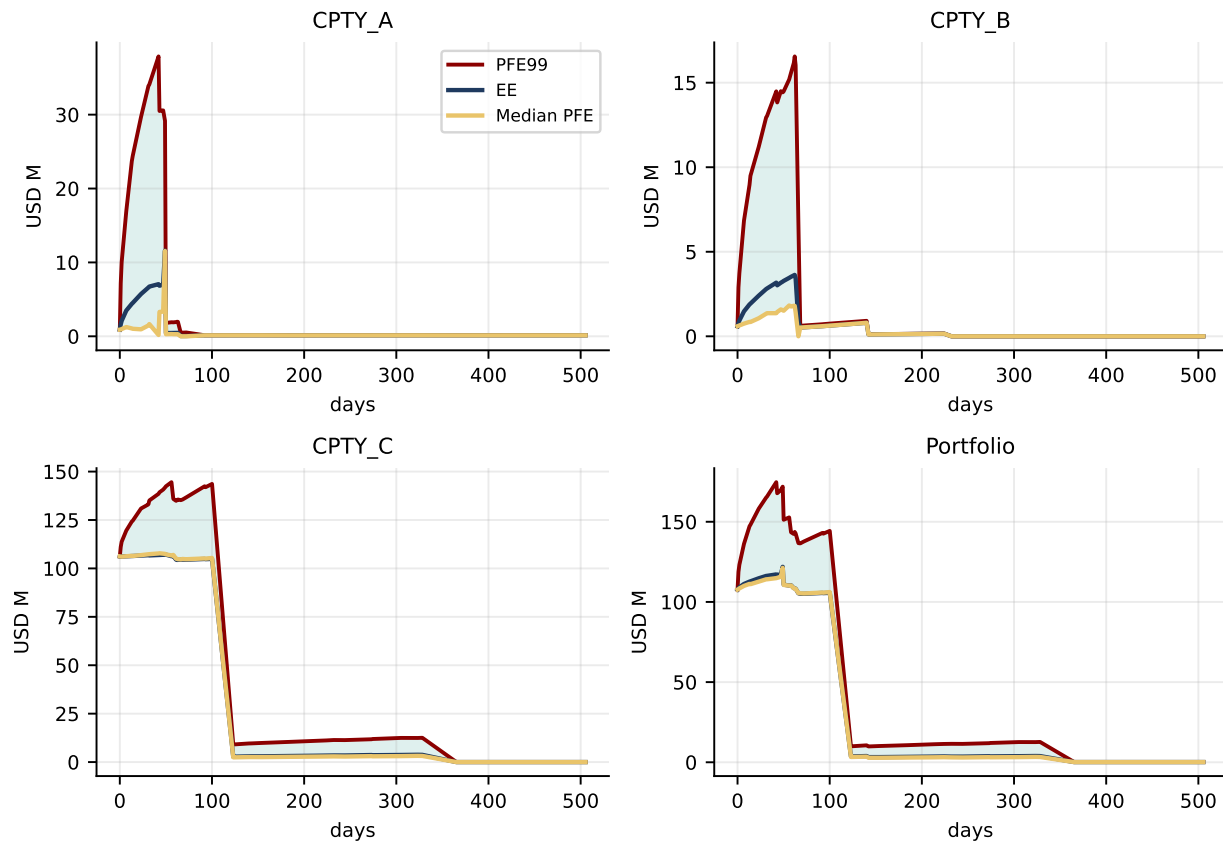


Figure 22: EE, median PFE and PFE99 for each counterparty and the portfolio. The shaded band runs from the median PFE to PFE99.

Counterparty	EE(0)	Peak EE	Peak median PFE	MPE (peak PFE99)	MPE date
CPTY_A	\$0.9M	\$11.6M	\$11.6M	\$37.9M	2026-10-09

Counterparty	EE(0)	Peak EE	Peak median PFE	MPE (peak PFE99)	MPE date
CPTY_B	\$0.6M	\$3.6M	\$1.8M	\$16.6M	2026-10-29
CPTY_C	\$106.2M	\$107.1M	\$107.8M	\$144.5M	2026-10-23
Portfolio	\$107.7M	\$121.9M	\$121.3M	\$174.7M	2026-10-09

Reading the shape. Exposure peaks soon after today and then falls steeply as trades mature. By the end of December 2026 the portfolio EE has fallen by more than 90%; the remainder is the Bond TRS BTRS_0001 and the year-long EQTRS_0008. So almost all counterparty credit risk sits in the next four months and monitoring should be concentrated there.

EE, median PFE and PFE99. The median PFE is far below EE for the counterparties with more than half of scenarios at zero exposure (CPTY_B especially): EE is pulled up by the right tail, the median PFE describes the typical outcome, and PFE99 describes the adverse tail. Reporting all three prevents any single statistic from misleading.

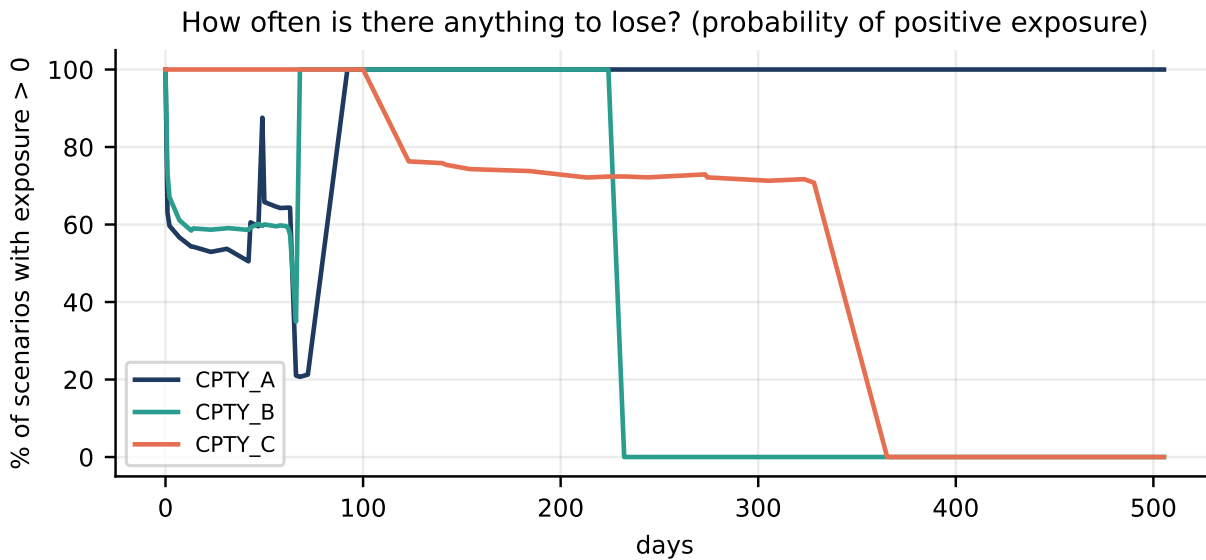


Figure 23: Probability that a counterparty has any positive exposure at all. Where this is far below 100%, the median PFE is zero even though EE and PFE99 are large.

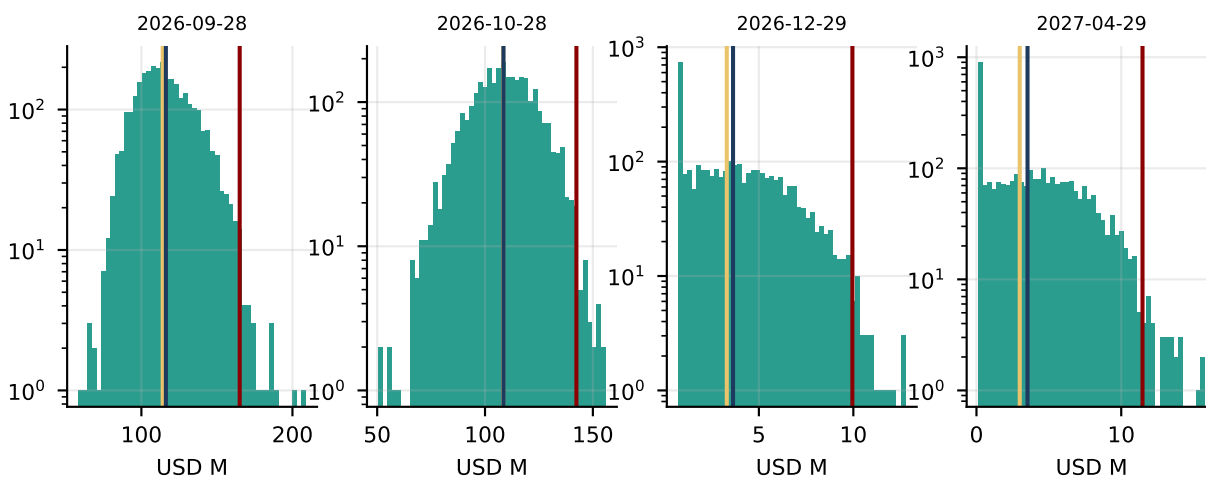


Figure 24: Portfolio exposure histograms (log count) at four dates with median (gold), EE (navy) and PFE99 (red). The right tail lengthens with time.

7.3 Where does the risk come from?

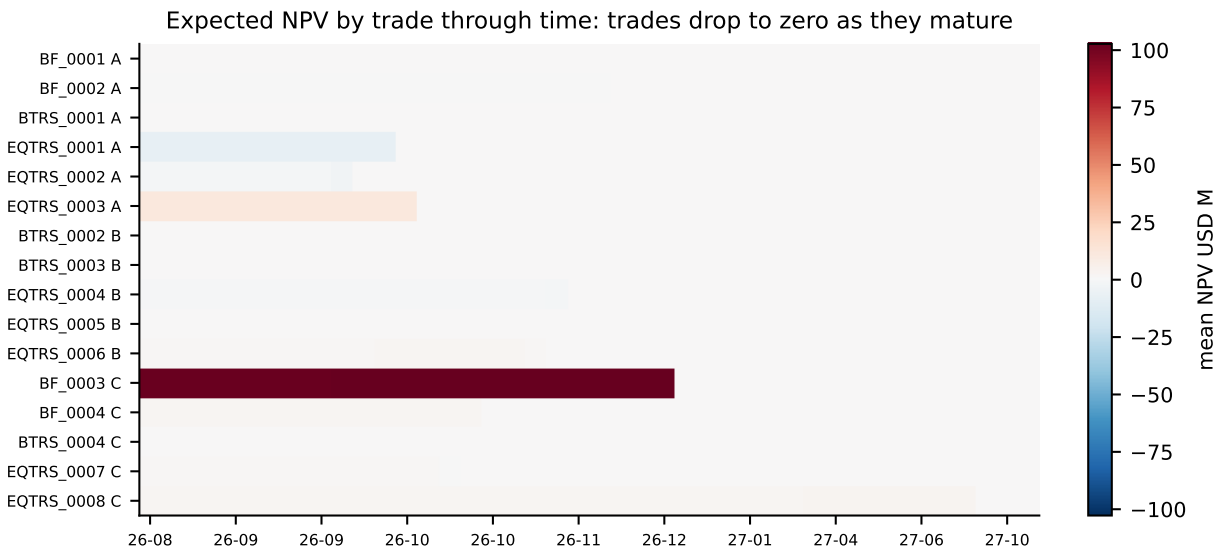


Figure 25: Expected NPV by trade at each reporting date. CPTY_C's BF_0003 dominates the picture and disappears at its 2026-12-06 settlement; a few equity TRS contribute negative expected NPV.

Concentration. CPTY_C accounts for 83% of the portfolio's peak PFE99 and 99% of today's exposure, and almost all of it comes from one trade, BF_0003 (500M notional, short forward struck at 100 on a long-dated 2.88% 2049 Treasury trading roughly 20 points below par, so we are owed the difference). This is concentration risk rather than diversified counterparty risk; a limit or collateral on that single trade would move the portfolio number more than any modelling choice in this report.

7.4 What if the counterparty posts margin? (MPOR-shifted exposure)

The real book is treated as **uncollateralized** because trade_data carries no CSA terms. For a collateralized counterparty, default does not mean instant close-out: there is a Margin Period of Risk (standard 10 business days) between the last collateral exchange and the actual replacement of the trades. The relevant exposure is what the position could gain in that window beyond the collateral held:

$$\begin{aligned}
 C(t) &= \max(V(t) - \text{threshold}, 0) && \text{collateral held at reporting date } t \\
 \text{Exposure}(t) &= \max(V(t + \text{MPOR}) - C(t), 0) && \text{same simulated path, MPOR days later}
 \end{aligned}$$

Implementation: an extra look-ahead node is inserted MPOR days after every reporting node on the same simulated path (no separate simulation). The demonstration below is explicitly hypothetical: full variation margin (threshold 0), MPOR 10 business days, 3,000 scenarios.

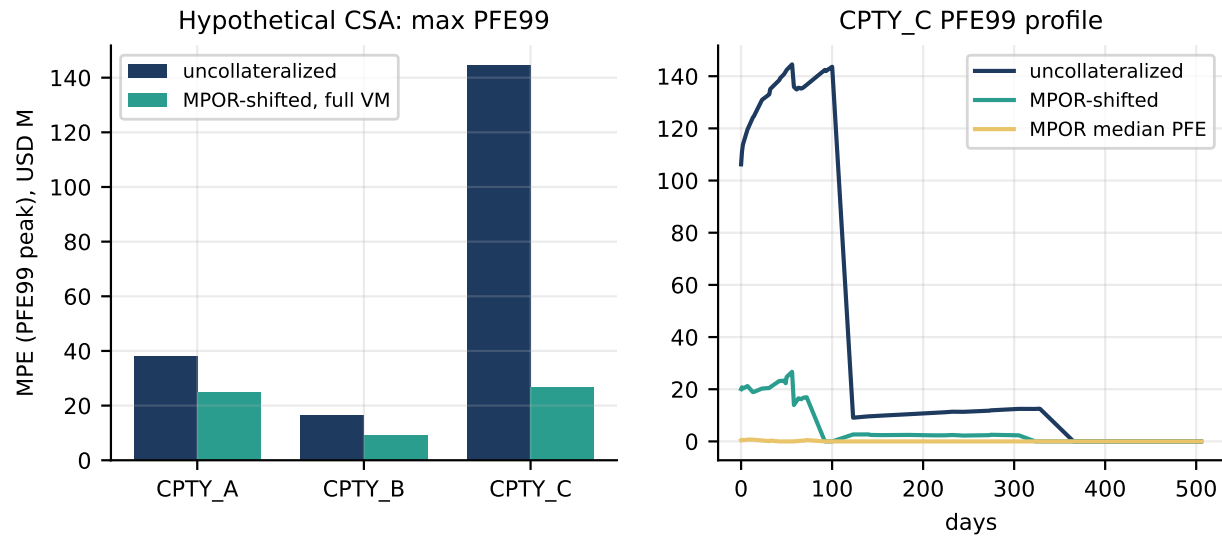


Figure 26: If a full-VM CSA existed, peak PFE99 would fall sharply, most of all for CPTY_C whose exposure is a large, already-visible MTM that margin would cover.

Counterparty	Uncollateralized MPE99	MPOR-shifted MPE99	Reduction
CPTY_A	\$37.9M	\$24.9M	34%
CPTY_B	\$16.6M	\$9.2M	45%
CPTY_C	\$144.5M	\$26.7M	82%

Relation to SIMM and Basel. The 10-day figure is the convention shared by both frameworks. Under the Basel counterparty credit risk rules (SA-CCR and the internal models method) the margin period of risk for a margined bilateral OTC netting set is at least 10 business days, 5 business days for centrally cleared trades and 20 business days for large or illiquid netting sets, and it lengthens after margin disputes. The ISDA Standard Initial Margin Model (SIMM) is a different object: a sensitivity-based methodology for the *initial margin* that the uncleared margin rules require counterparties to post, calibrated to a 99% one-tailed loss over a 10-day horizon using a stress period. Initial margin is collateral posted in addition to variation margin and held against precisely the close-out window described above.

Our engine does **not** implement SIMM and does not model initial margin. The calculation in this section is a variation-margin-only illustration: a threshold of zero removes any unsecured amount, but there is no initial margin buffer. If initial margin were posted it would absorb part of the 10-day move and reduce the exposure shown further; quantifying that requires computing SIMM sensitivities for each netting set, which we have not attempted. Two simplifications should also be noted: the look-ahead is exactly 10 business days (Monday to Friday, US federal holidays skipped; the SIFMA bond-market calendar differs marginally), and no minimum transfer amount is modelled.

Whether any trade is actually margined is the most important open question for Capitols: it changes the answer by roughly an order of magnitude for CPTY_C.

7.5 Full-rank correlation versus PCA factor model

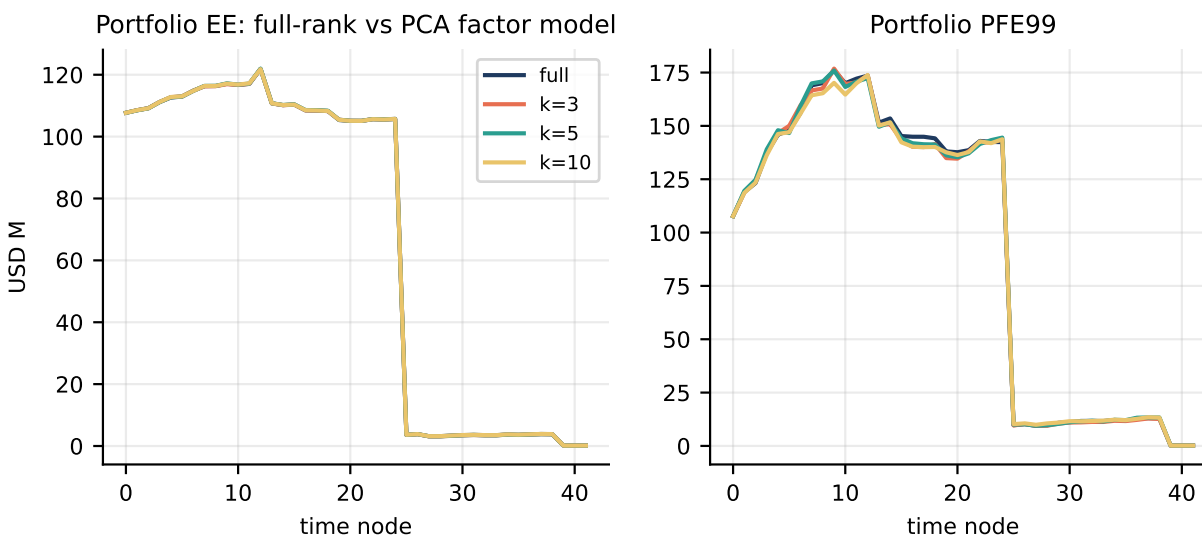


Figure 27: Same seed and method (1,500 scenarios) with the full Cholesky correlation versus a k-factor PCA model. Curves coincide closely from k=5.

Model	Peak portfolio EE	MPE (peak PFE99)	MPE difference vs full
full	\$121.8M	\$176.1M	-
k=3	\$121.9M	\$176.9M	+0.4%
k=5	\$122.0M	\$175.9M	-0.1%
k=10	\$121.8M	\$173.8M	-1.3%

This is a like-for-like comparison at one scenario count, so differences include sampling noise of a few tenths of a percent as well as the approximation itself. The factor model with about 5 factors reproduces the book's exposure closely, which supports it as a lower-dimensional, more robust alternative, but the full-rank default is kept because it is exact with respect to the estimated matrix. That five factors (only about half of the correlation matrix's total variance) suffice here is because this book's exposure is driven by the rate, a few large positions and FX, not by the fine correlation structure among all 37 names.

8 Credit valuation adjustment and SA-CVA capital

Sections 1 to 7 measure how much a counterparty could owe us. This section prices the risk that they fail to pay it (credit valuation adjustment, CVA) and computes the Basel regulatory capital held against the volatility of that CVA under the Standardised Approach (SA-CVA). Both reuse the exposure engine unchanged.

8.1 Definition

CVA is the expected loss from counterparty default, assuming the bank itself cannot default (unilateral CVA). Following Basel MAR50.32:

```

CVA = LGD * sum_i 0.5 * [ DEE(t_{i-1}) + DEE(t_i) ] * PD(t_{i-1}, t_i)
DEE(t) = E[ D(t) * max(V(t), 0) ]          D = pathwise risk-free discount factor
PD(t_{i-1}, t_i) = exp(-s_{i-1} t_{i-1}/LGD) - exp(-s_i t_i/LGD)    (market-implied, from spreads)

```

The expected discounted exposure DEE is the exposure profile of Section 7 with each scenario discounted along its own simulated short rate. The default probability is implied by a credit spread s through the credit-triangle relation with a loss given default LGD of 60% (40% recovery, the senior unsecured convention). Exposure and default are assumed independent, so wrong-way risk is excluded, and the book is treated as uncollateralized because no margin terms are available.

8.2 Credit inputs: rating proxies from bond spreads

CDS spreads for the counterparties are not available, and the counterparties are anonymised. MAR50.32(3) permits an illiquid counterparty's spread to be proxied from liquid peers by credit quality, industry and region. We proxy by credit quality using public bond-market data: the ICE BofA US corporate option-adjusted spread indices by rating (FRED), with the maturity shape taken from the ICE BofA corporate index by maturity bucket, scaled to each rating. The rating-level spread is one number per rating, so the same maturity shape is used for all ratings (a disclosed simplification).

Sources. Rating spreads: FRED series BAMLC0A1CAAA (AAA), BAMLC0A2CAA (AA), BAMLC0A3CA (A), BAMLC0A4CBBB (BBB), BAMLC0A1HYBB (BB) and BAMLC0A2HYB (B), ICE BofA US option-adjusted spreads. Maturity shape: BAMLC1A0C13Y, BAMLC2A0C35Y, BAMLC3A0C57Y and BAMLC4A0C710Y relative to BAMLC0A0CM (all US corporate). Basel parameters: Bank for International Settlements, Targeted revisions to the credit valuation adjustment risk framework (July 2020), MAR50. Equity bucket attributes (sector, country, market capitalisation): yfinance.

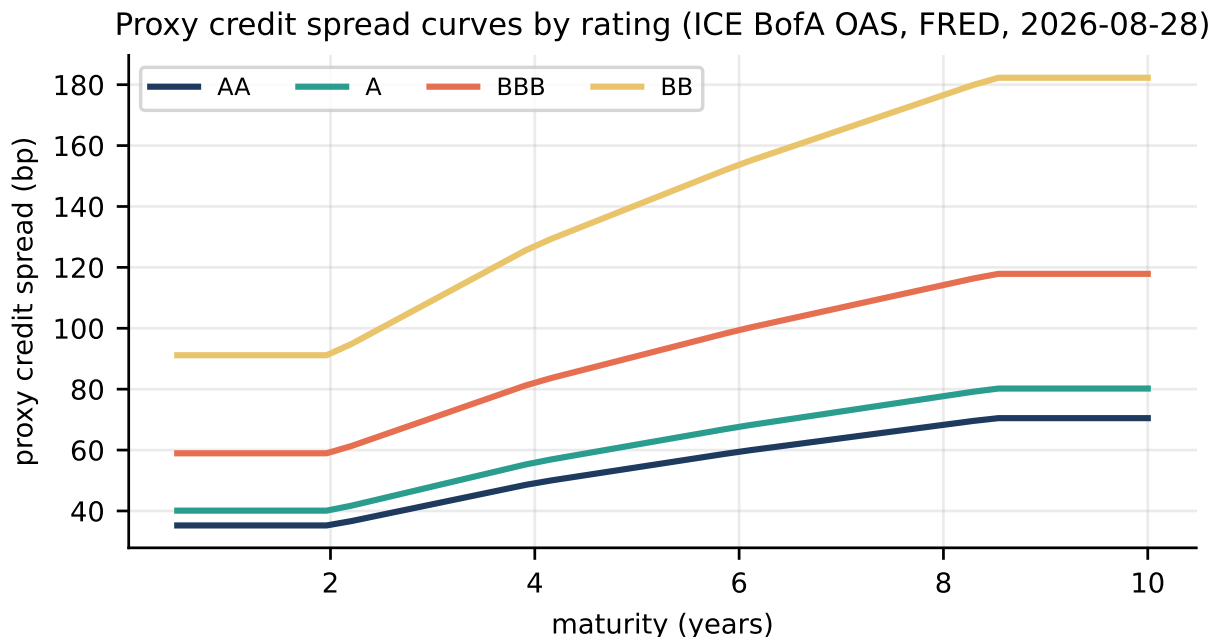


Figure 28: Proxy spread curves by rating on the valuation date. The BBB curve used for the counterparties runs from about 59 bp at 6 months to about 118 bp at 10 years.

Counterparty	Assumed rating	Assumed sector	Basis
CPTY_A	BBB	Financials (SA-CVA bucket 2)	Assumption: unrated counterparty proxied at BBB
CPTY_B	BBB	Financials (SA-CVA bucket 2)	Assumption
CPTY_C	BBB	Financials (SA-CVA bucket 2)	Assumption

The counterparty identities and ratings were not provided, so these are explicit assumptions. Because they matter more than any modelling choice, Section 8.3 reports CVA across the whole rating range so the reader can substitute the correct rating.

8.3 Results: CVA

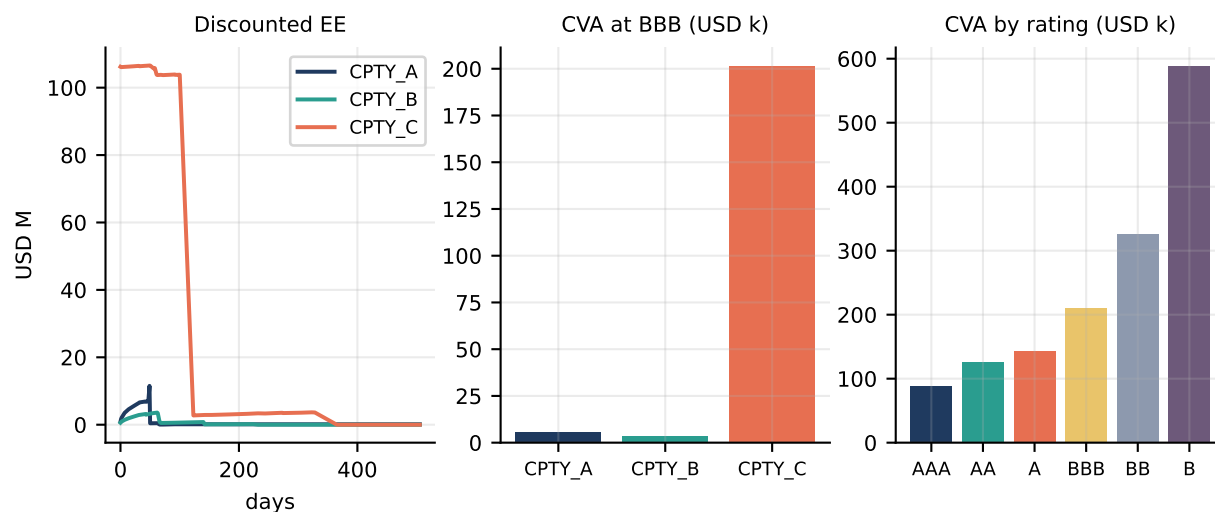


Figure 29: Left: expected discounted exposure by counterparty. Middle: CVA by counterparty at the BBB proxy. Right: portfolio CVA if all counterparties had the rating shown.

Counterparty	CVA (USD)	Share of total
CPTY_A	5,403	3%
CPTY_B	3,701	2%
CPTY_C	201,291	96%
Portfolio	210,394	100%

Rating	Portfolio CVA (USD)	Multiple of BBB
AAA	89,022	0.42x
AA	125,894	0.60x
A	143,237	0.68x
BBB	210,394	1.00x
BB	325,032	1.54x
B	588,058	2.80x

CVA is concentrated where the exposure is: CPTY_C carries 96% of it, as it carries almost all of the exposure. CVA scales approximately linearly with the credit spread, so the rating assumption moves the result by a factor of about 2.6 between AA and BB. These figures use 1,000 scenarios (Latin Hypercube) with common random numbers across the sensitivity runs of Section 8.4.

8.4 Basel SA-CVA capital

SA-CVA (MAR50.27 to 50.77) sets capital from the sensitivity of regulatory CVA to market risk factors. It is an adaptation of the market-risk standardised approach and needs supervisory approval, a CVA desk and monthly sensitivity calculation (MAR50.30); the figures here are therefore indicative of the requirement, not a filed number. For each risk factor k the CVA sensitivity s_k is computed by bump-and-reprice with common random numbers, converted to a weighted sensitivity $WS_k = RW_k \times s_k$, and aggregated by bucket and risk class:

```
K_b = sqrt( sum_k WS_k^2 + sum_{k!=l} rho_kl WS_k WS_l )      S_b = clip( sum_k WS_k, -K_b, +K_b )
K   = m_CVA * sqrt( sum_b K_b^2 + sum_{b!=c} gamma_bc S_b S_c )
Capital = sum over classes of ( K_delta + K_vega );   RWA = 12.5 * Capital
```

Risk classes present in the book, the shift used for each sensitivity and the Basel parameters applied:

Risk class	Risk factors and shift	Risk weight	Correlation
Interest rate delta (USD)	USD risk-free yield at 1, 2, 5, 10, 30y; +1bp with triangular weights	1.11%, 0.93%, 0.74%, 0.74%, 0.74%	Tenor matrix 31-91% (Table 4)
Interest rate vega (USD)	All USD rate volatilities, +1% relative	100%	Single factor
FX delta (USDJPY)	USDJPY spot, +1% relative	11%	Single factor; cross-bucket 0.6
FX vega	USDJPY volatility, +1% relative	100%	Single factor
Counterparty credit spread	Each counterparty at 0.5, 1, 3, 5, 10y; +1bp	5% (financials, investment grade)	Tenor 0.9 x name 0.5 (unrelated)
Equity delta	All names in an equity bucket, +1% relative spot	30-55% by bucket	Cross-bucket 0.15
Equity vega	All volatilities in a bucket, +1% relative	78% large cap, 100% small cap	Cross-bucket 0.15
Reference credit, commodity	None: no such drivers of exposure	-	-

Equity names are assigned to Basel buckets by size (market capitalisation of at least USD 2 billion), region (advanced versus emerging economy) and sector, using yfinance data: bucket 1: 1 names, bucket 4: 2 names, bucket 5: 13 names, bucket 6: 6 names, bucket 7: 2 names, bucket 8: 13 names.

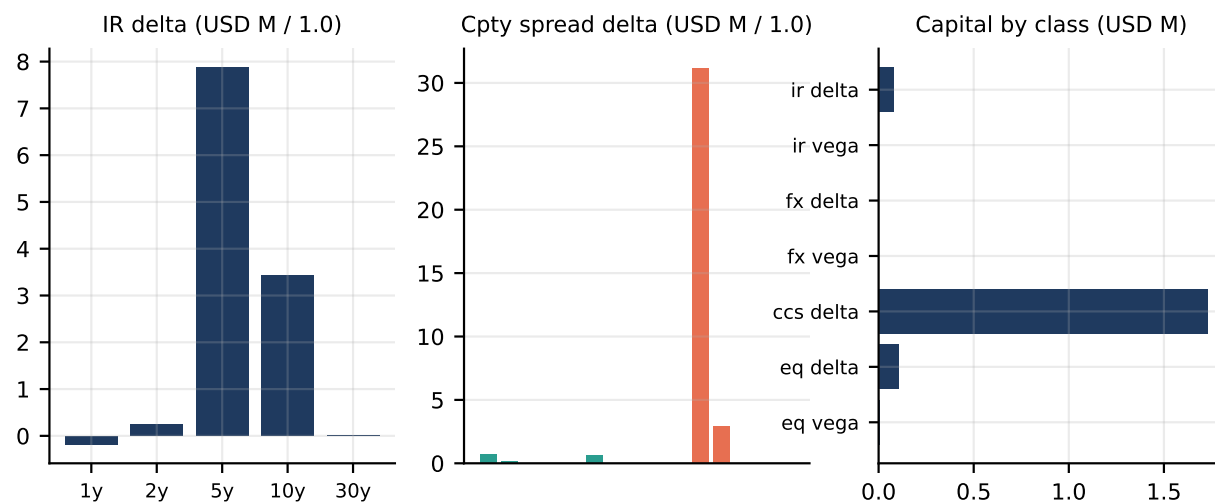


Figure 30: Left: CVA sensitivity to a 1bp rise in USD yields at each SA-CVA tenor. Middle: sensitivity of CVA to the counterparty credit spreads (three counterparties by five tenors). Right: SA-CVA capital by risk class.

Risk class	Capital K (USD)
ir delta	82,625
ir vega	89
fx delta	2,479
fx vega	0
ccs delta	1,729,841
eq delta	105,510
eq vega	6,637
Total SA-CVA capital requirement ($m_CVA = 1$)	1,927,180
Risk-weighted assets (12.5 x capital)	24,089,749
Total if the pre-2023 multiplier $m_CVA = 1.25$ applied	2,408,975

The requirement is dominated by ccs delta (90% of the total), as is typical: a 5% risk weight on credit spreads is large relative to a 100 bp spread level, so the capital charge is a multiple of the CVA itself. Interest rate, FX and equity classes are comparatively small. The multiplier m_CVA is 1 in the framework in force from 2023 (it was 1.25 in the original 2017 text); both are shown.

8.5 Assumptions and limitations of the CVA work

- **Counterparty ratings are assumed** (BBB financials). CVA and the credit-spread capital scale with this assumption; Section 8.3 shows the range.
- **Spreads are bond-index proxies**, not CDS: the maturity shape is common to all ratings, and the indices are US corporate spreads regardless of counterparty region or sector.
- **Unilateral, independent, uncollateralized.** No DVA, no wrong-way risk (exposure and default dependence) and no margin, which MAR50.32 allows to be recognised only where terms are known.
- **Interest rate risk is USD only:** the JPY rate is not simulated, so JPY rate sensitivity is captured through the USD curve; inflation risk factors are omitted (no inflation exposure).
- **Sensitivities are one-sided bumps** from 1,000 Latin Hypercube scenarios with common random numbers, not adjoint sensitivities; vega shifts apply to the volatilities driving the simulated paths (the trades contain no options, so there are no option-pricing volatilities).
- **Parameters** are transcribed from the BIS text (July 2020 revisions); the vega risk weights follow the 100% and 78% values in that text. Regulatory use requires supervisory approval and validation of the sensitivity calculation.
- The Basel Basic Approach (BA-CVA) is not computed; it needs only counterparty exposure and maturity and is the natural fallback if SA-CVA approval is not held.

9 Sensitivities (Greeks)

Capitolis asked for exposure profiles together with efficient Greeks across scenarios, aggregated by netting set. This section delivers both levels: (i) the Greeks of the book's value today, by trade and netting set, and (ii) the sensitivities of the exposure measures themselves, EE, median PFE and PFE99, at every simulation date. Every number comes from the same pricers and simulated paths as Section 7.

9.1 Definitions and method

- **Delta** to an equity or to USDJPY: change per +1% relative move of the spot. **Gamma**: change in that delta, $[M(+1\%) - 2 M + M(-1\%)]$. **DV01**: change per +1bp in USD zero rates, parallel and bucketed at 0.25, 0.5, 1, 2, 3, 5, 10 and 30 years (triangular weights that sum exactly to the parallel shift). **Vega**: change per +1% relative shift of the volatility of the USD rate, of USDJPY and of all equities.
- **Bump-and-reprice with common random numbers** (the same Latin Hypercube draws in base and bumped runs): the difference is then sensitivity, not Monte Carlo noise. It works on the pricers as a black box, and it applies equally to quantile measures such as PFE99, which have no simple pathwise derivative. Its accuracy compared with the pathwise estimator is shown in Section 11.6.
- **Efficiency**. A bump of an equity or of USDJPY rescales every simulated path exactly (the GBM drift does not depend on the spot level), so no re-simulation is needed: the existing paths are reused with the spot multiplied and only the trades holding that factor are repriced, all bumps in one process pool. Rate and volatility bumps change the paths themselves, so they are re-simulated with the same random numbers.
- Exposure Greeks are computed at $N = 2,000$ scenarios (Latin Hypercube), for the uncollateralized book, on the pillar date grid.

Sign convention: Capitolis is strictly the seller on these trades, and the equity swaps are of type pay-equity (we pay the equity return and receive funding), so equity deltas are negative: our value rises when the shares fall.

9.2 Book Greeks today

Netting set	NPV	Equity delta (per +1%)	Equity gamma	FX delta (per +1%)	DV01 (per +1bp)	Rate gamma
CPTY_A	897,526	-2,280,302	0	0	31,258	-16
CPTY_B	586,826	-966,846	0	420,465	-3,752	6
CPTY_C	106,202,202	-1,231,141	0	0	637,630	-1,300
Portfolio	107,686,554	-4,478,289	0	420,465	665,137	-1,310

All values in USD. The equity and bond trades are linear in the equity spot, so equity and FX gamma are zero to rounding; rate gamma is small. The DV01 is dominated by the CPTY_C bond forward. The largest deltas and the bucketed DV01s:

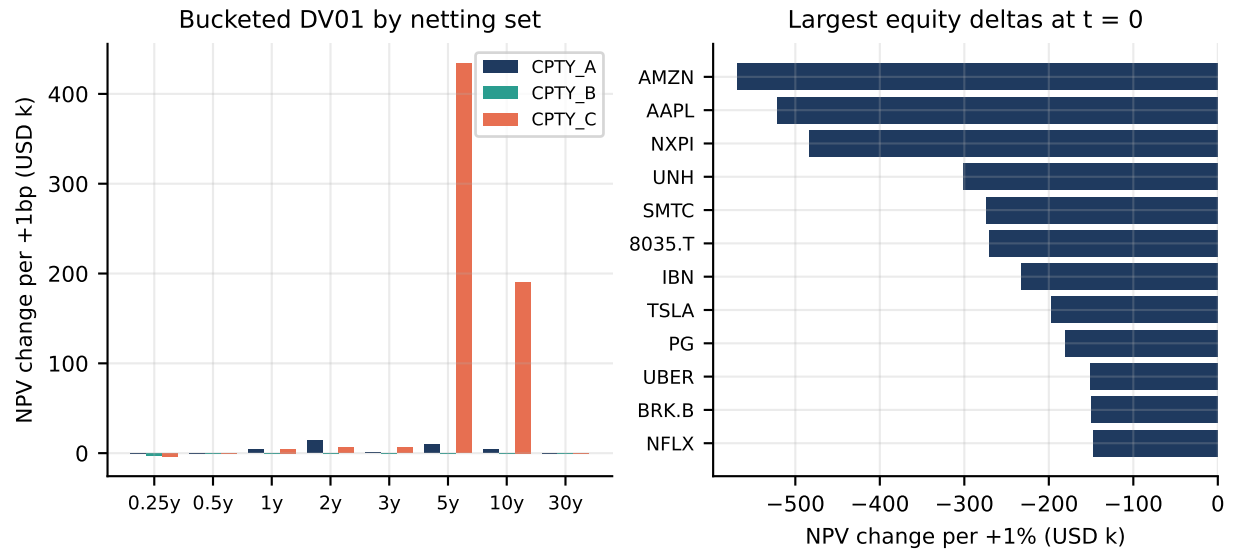


Figure 31: Left: DV01 by netting set and tenor. Right: the twelve largest equity deltas at $t = 0$.

Netting set	0.25y	0.5y	1y	2y	3y	5y	10y	30y
CPTY_A	-565	83	3,715	13,865	399	9,692	4,069	0
CPTY_B	-2,649	-905	-72	-1	-1	-86	-38	0
CPTY_C	-3,776	-102	4,667	6,291	6,240	433,804	190,506	0

Bucketed DV01 (USD per +1bp), summing to the parallel DV01 above. A check that the equity delta is right: for EQTRS_0003 the delta to its largest name equals minus the share count times the spot, i.e. exactly -212,269 USD per \$1, as it must for a linear payoff (Section 9.5).

9.3 Sensitivities of the exposure measures

These are the Greeks that matter for counterparty credit risk: how EE, median PFE and PFE99 change when a market factor moves. The figure shows the portfolio profiles for the three main factors, and the table gives the sensitivities at the date of peak PFE99 (2026-10-09), by netting set.

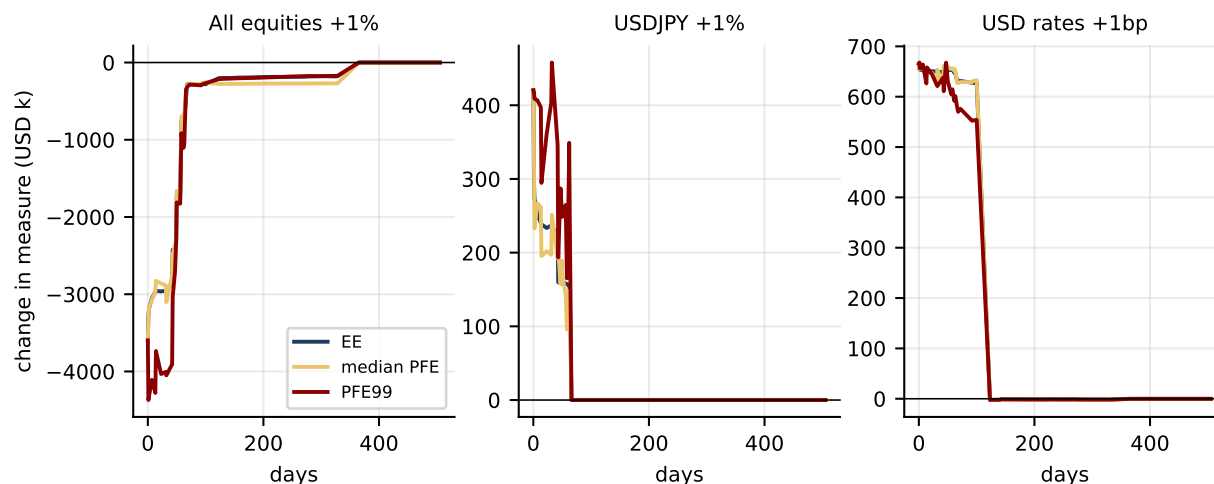


Figure 32: Change in portfolio EE, median PFE and PFE99 through time for +1% on all equities, +1% on USDJPY and +1bp on USD rates.

Factor shift	Measure	CPTY_A	CPTY_B	CPTY_C	Portfolio
All equities +1%	EE	-1,095,935	-539,348	-1,233,470	-2,868,753
	Median PFE	-1,363,675	-964,461	-1,244,428	-2,759,673
	PFE99	-1,894,675	-829,702	-1,044,390	-3,907,784
USDJPY +1%	EE	0	226,832	0	226,832
	Median PFE	0	412,132	0	192,426
	PFE99	0	314,078	0	345,892
USD rates +1bp	EE	14,839	-1,951	635,801	648,689
	Median PFE	29,137	-3,393	627,855	624,517
	PFE99	28,938	-3,311	610,815	637,953
Rate volatility +1%	EE	157	18	-119	55
	Median PFE	-3,667	176	1,745	26,048
	PFE99	8,845	744	126,540	125,471
FX volatility +1%	EE	0	0	0	0
	Median PFE	0	0	0	0
	PFE99	0	0	0	0
Equity volatility +1%	EE	73,227	25,802	-541	98,487
	Median PFE	23,427	5,040	-27,261	40,892
	PFE99	362,225	118,917	171,905	535,707
All-equity gamma (+1%)	EE	119,200	55,173	0	174,373
	PFE99	-0	-0	0	-0
USDJPY gamma (+1%)	EE	0	6,154	0	6,154
	PFE99	0	-6,399	0	-6,918
Rate gamma (+1bp)	EE	21	5	-1,276	-1,250
	PFE99	-14	5	-1,223	-1,236

Change in the measure, in USD, at 2026-10-09. Delta and vega are one-sided shifts read as change per unit shift; gamma is the second difference.

Reading the table. The equity delta of exposure is negative (EE falls by about \$2,869k per +1% on all equities and PFE99 by \$3,908k): the swaps are pay-equity, so our claim on the counterparty shrinks when shares rise, and the adverse scenarios for credit exposure are equity falls. The USD rate DV01 of PFE99 is positive (\$638k per +1bp), reflecting the funding legs and bond forwards. The equity and rate gammas of the exposure are positive, as expected for $\max(V, 0)$ (a call on the netted value), and small relative to the deltas at a +1% shift.

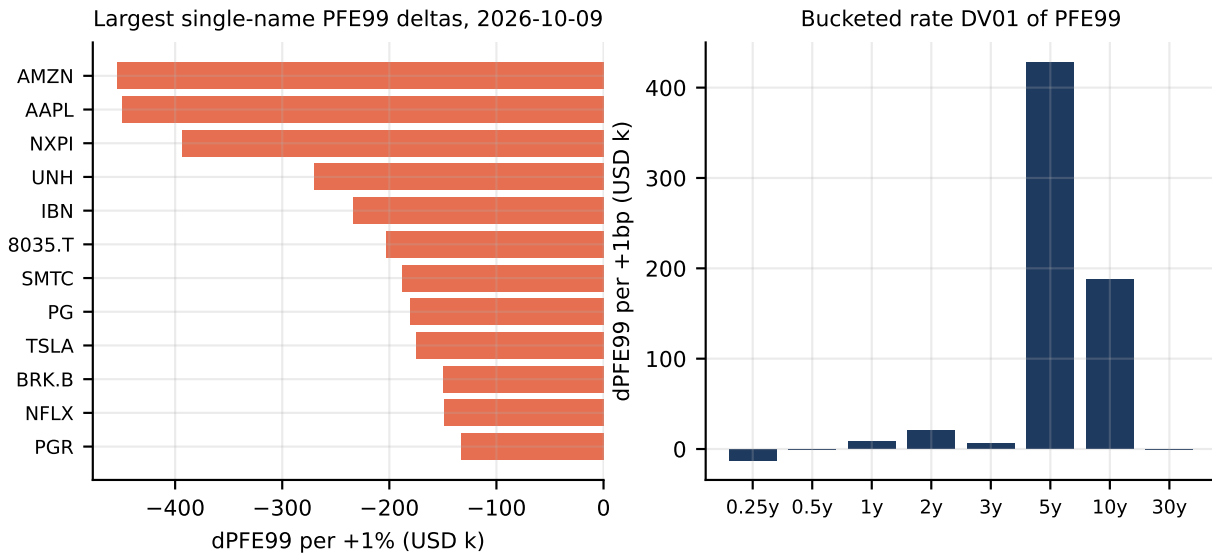


Figure 33: Left: the twelve largest single-name deltas of portfolio PFE99 at the peak date. Right: bucketed USD rate DV01 of PFE99 at the same date.

Single-name deltas of PFE99 show which positions drive the tail; because a quantile is not additive across names, the single-name deltas sum only approximately to the all-equity delta (Section 9.5).

9.4 Efficiency

Step (timed at N = 300)	Cost	Comment
t = 0 book Greeks	2.2 s	Central differences on the pricers: every trade, 37 equities, FX, DV01 parallel and 8 buckets
Base run (paths and full repricing)	115 s	16 trades, all dates, 8 worker processes (includes process start-up)
78 equity and FX bumps	110 s	No re-simulation; only the trades holding the factor are repriced; one process pool. Re-simulating each would cost about 10,317 s (94x more)
One rate or volatility re-simulation	132 s	12 are needed: parallel up and down, 8 buckets, three vega bumps

Timings were measured separately at N = 300 because the wall-clock of the N = 2,000 production run was distorted by the machine sleeping during the overnight job; costs scale about linearly with N once process start-up is amortised.

The equity and FX Greeks, which are 37 names plus FX, each with an up and a down bump, cost a small fraction of what naive bump-and-resimulate would. This is the main efficiency gain: it exploits the structure of the model (exact rescaling of GBM paths) rather than approximating anything.

9.5 Validation of the Greeks

Check	Result
t = 0 equity delta against the analytic value (shares x spot x 1%, signed by trade direction), all 41 trade-name pairs	maximum relative error 5.5e-04
Bump-size stability of the all-equity EE delta at the peak date: 0.5%, 1%, 2% bumps (USD k per +1%)	-2,870, -2,869, -2,869 (spread 0.03%)
Additivity: sum of the 37 single-name EE deltas against the all-equity delta (maximum over all dates after t = 0)	1.34% of the peak all-equity delta

Check	Result
The same at $t = 0$ itself	25%: at $t = 0$ the exposure $\max(V, 0)$ is deterministic and has a kink, and a 1% bump of one name can cross it (CPTY_A, CPTY_B), so single-name and joint finite differences need not add there
Sum of single-name PFE99 deltas against the all-equity PFE99 delta at the peak date (quantiles are not additive)	-3,908 against -3,908 USD k
Common random numbers against independent draws: standard deviation of the EE delta estimate over 4 repeats at $N = 300$	33 USD k against 472 USD k (14x lower)

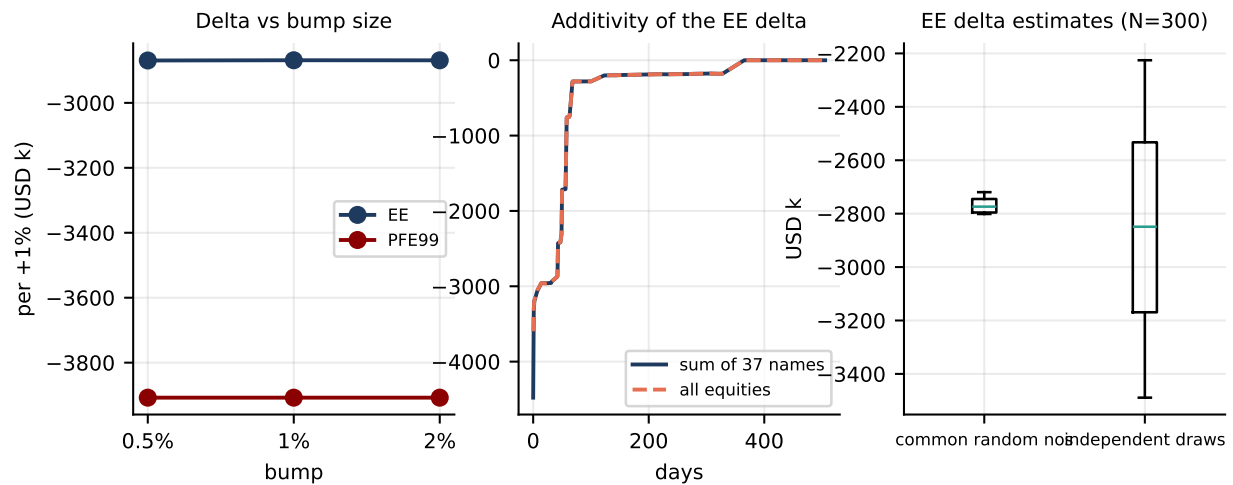


Figure 34: Left: delta of EE and PFE99 against bump size. Middle: the sum of the 37 single-name EE deltas equals the all-equity delta through time. Right: estimates of the EE delta with common random numbers and with independent draws.

9.6 Limitations

- Sensitivities are finite differences with a 1% (equity, FX, volatility) or 1bp (rates) shift. The bump-size check above shows the results are stable to a factor of four in bump size.
- Quantile Greeks (PFE99) carry more estimation noise than EE even with common random numbers; the reported figures use 2,000 scenarios.
- Interest rate Greeks are for the USD curve only: the JPY rate is not simulated, so JPY rate risk enters through the USD curve. There is no inflation, credit-spread or dividend Greek for exposure (credit-spread and vega sensitivities of CVA are in Section 8).
- Volatility bumps are parallel across each factor class (all equity vols together); there is no vol-surface or per-name vega. No cross-gamma is computed, and theta is represented by the exposure profile itself.
- Exposure Greeks are for the uncollateralized book at 99%; a collateralized book would have different (much smaller) sensitivities.

10 Every modelling choice, and what we rejected

Topic	Choice	Alternatives considered	Reason
Measure	Risk-neutral	Real-world drifts	No unobservable risk premia; verified martingale; horizon short
Confidence	PFE99 and median PFE (with EE)	Other percentiles	99% is the specified target; 99.9% needs far more paths for little added insight
USD rates	Hull-White 1F	CIR, BK, LMM, constant	Exact curve fit, analytic bonds, negative-rate capable, few parameters
Equities/FX	Correlated GBM	Heston, SABR, jumps	No option surfaces; matches lognormal vol convention
Volatility	3y realised	Implied vols	No single-name options data; documented proxy
Correlation	One static matrix, Cholesky	Time-varying (DCC), stressed, factor	Data supports one matrix; factor model offered as alternative
Equity factor model	Optional PCA (k=5)	Full rank only	Robustness, dimension reduction; approximation disclosed
Dates	Pillar dates + trade event dates	Monthly grid	Industry convention, denser where risk moves, exact cash-flow dates
Sampling	Latin Hypercube	Pseudo-random, antithetic, moment-matched, Sobol	Best measured error on both test and real engine
Scenario count	5,000 (reporting); 1,000 (iteration); 10,000 (sign-off)	30,000 pool	Diminishing returns beyond 10-15k; model risk dominates
Mean reversion a	Swaption-based (USD 0.0167)	Futures proxy (0.0458), textbook 0.03	Industry-standard instrument
JPY rates	Real JPY Hull-White factor (built), not yet wired	Constant differential only	Negative-rate capable; real TONA sigma; correlation calibrated ~0
Collateral	Uncollateralized default; MPOR-shift option	Assume full VM	No CSA data; hypothetical shown separately
Pricing	Reuse validated pricer library	Vectorised reimplementation	Correctness first; speed via multiprocessing

11 Validation: how we know the engine is right

Each check isolates one layer (paths, pricing, calibration, aggregation) so a failure points at where to look.

11.1 Self-consistency at $t = 0$

Simulated $EE(0)$ must equal a direct current-exposure calculation from the identical calibrated snapshot: matches to 0.0000% (after fixing two comparison-object mismatches, Section 10).

11.2 Martingale / no-arbitrage test

Under the risk-neutral measure discounted asset prices must have no drift. With 8,000 paths: the bank-account check $E[\exp(-\int r)] = P(0,T)$ holds to a maximum relative difference of 0.73 bp across nodes (a small, understood trapezoid-integration effect of the coarse monthly grid), and the discounted-equity ‘gains process’ for six names has all $|z|$ below 1.4. This validates the drift terms in the engine itself, independent of any trade pricing.

11.3 Parametric (delta-normal) VaR benchmark

An independent standard method: portfolio dollar-deltas by bump-and-reprice with the same volatilities and correlations as the simulation, combined into a delta-normal VaR at the first simulated node (one day, the overnight pillar). The 99% delta-normal VaR is \$11.4M against \$11.7M from the Monte Carlo P&L distribution (ratio 0.98; accepted range 0.3-1.3). At this short horizon the book is close to linear in the risk factors, so close agreement is expected; a large discrepancy would have indicated a sign or scaling error in the deltas or the covariance.

11.4 Sensitivity (stress) test

Directional test with common random numbers (800 scenarios, same seed in base and bumped runs). Base MPE (PFE99) is \$174.0M. Raising equity volatility by 50% increases it to \$189.8M (+9.1%); a +100bp parallel shift of the USD curve to \$230.5M (+32.5%), consistent with the USD rate being the largest single delta; and raising Hull-White sigma by 50% to \$183.4M (+5.4%). All three moved exposure in the economically required direction. The magnitudes also show the scale of model risk: an equity volatility error of 50% matters about ten times more than the sampling error at the recommended path count.

11.5 Statistical and unit tests

82 automated tests pass. They cover: Hull-White reproducing the curve at $t=0$ to $1e-9$; the GBM martingale property; the $1/\sqrt{N}$ error law; antithetic variance reduction; Cholesky recovering a target correlation; MPOR look-ahead spacing and formula; pillar dates and forced event dates; negative-rate behaviour of Hull-White; the PCA factor model (exact at full rank, monotone error, variance preservation); the BOJ TONA and MOF JGB loaders and the empirical JPY findings; the tail-convergence result; and median PFE, CVA and the SA-CVA aggregation, the credit-spread proxy curves, and the Greeks machinery (bump helpers, $t=0$ Greeks against analytic values, exposure measures).

11.6 Greeks (sensitivities)

Three delta estimators were compared on a call option: pathwise, bump-and-reprice with **common random numbers** (same draws in base and bumped runs) and bump-and-reprice with independent draws.

Method	Bias	Std of estimate	Time/trial
Pathwise	+0.000085	0.003771	0.470 ms
Bump, common random numbers	+0.000033	0.003742	0.327 ms
Bump, independent random numbers	-0.003502	0.096885	0.850 ms

Common random numbers make bump-and-reprice statistically indistinguishable from the pathwise estimator (26x lower noise than independent draws), and it works on any pricer as a black box, so it is the method that generalises to the real book. On the real book, the exact $t=0$ delta of EQTRS_0003's NPV to its largest position is -\$212,269 per \$1 move, exactly equal to the share count (a linear-payoff consistency check).

12 Defects identified and resolved

Each defect was identified by checking against an independent source, a hand calculation or a full end-to-end run.

#	Bug	How it was caught / fix
1	BRK.B vs BRK-B ticker silently failed in two functions	Missing prices; fixed in both current and historical fetch
2	SOFR curve: recently-expired serial futures wrapped a decade forward, creating a multi-year gap	Cross-check against Treasury.gov par curve (73bp vs smooth 56bp at 10Y); fixed
3	The fix for #2 over-corrected and dropped genuine far-dated live contracts (e.g. SR3H0, March 2030)	Found while calibrating Hull-White; fixed with a 400-day plausibility threshold; curve coverage 3.6y -> 6.3y
4	US and Tokyo equity histories joined by timestamp gave a ~70% blank table	Join by calendar date instead
5	Stale cached exposure file gave false 7-15% 'discrepancies'	Root cause: real overnight moves (one name has 76% vol); check now uses the in-memory snapshot
6	Residual 0.02-0.03% self-check gap	Different curve object; both sides now use the identical exact analytic curve: 0.0000%
7	Tail-convergence study initially run at 99.9% rather than the specified 99%	Re-run at 99%; tests and documentation updated
8	JPY mean-reversion fit gave a negative a	Diagnosed as a real structural property (three sources); lower bound a = 0.001 used

13 Assumptions and limitations

- **Uncollateralized and no CSA data.** If margin exists, results change by up to an order of magnitude (Section 7.4).
- **Volatility is a 3-year realised proxy**, not implied. Regime shifts and skew are not captured.
- **One static correlation matrix** from 613 days; correlations tend to rise in stress and are not stressed here.
- **GBM underestimates fat tails**, most relevant for PFE99 on high-vol names.
- **Single-factor USD rates:** no curve twists independent of the level.
- **Risk-neutral drift:** not a real-world forecast; regulatory PFE may need physical drift.
- **JPY:** mean reversion is at its lower bound (the data give a negative value) and the JPY rate does not yet drive JPY equity/FX drift (two trades).
- **Model risk vs sampling risk:** above a few thousand scenarios the uncertainty is in the models, not the random numbers.
- **Margin modelling is partial:** the collateral illustration is variation-margin only, uses a 10-business-day window on a US federal holiday calendar, and has no initial margin (SIMM) or minimum transfer amount.
- **No wrong-way risk or credit dynamics of the counterparty itself;** this is exposure, not a loss estimate (that needs PD and LGD).
- **Bloomberg data is a single 2026-08-31 snapshot** (three days after the 2026-08-28 market data); acceptable for shape and level, disclosed.

14 Conclusions and recommendations

- The engine is validated (t=0 self-check, martingale, VaR benchmark, stress test, 82 tests) and reproducible with a single command per stage.
- The uncollateralized book has peak portfolio PFE99 of \$174.7M at 2026-10-09, versus EE of \$121.9M and median PFE of \$121.3M; the spread between these three is the point of reporting all of them.
- Risk is short-dated (about four months) and concentrated (one trade, one counterparty).
- CVA on the uncollateralized book is about \$210k at a BBB proxy (\$126k at AA to \$325k at BB); the SA-CVA requirement is \$1.93M (RWA \$24.1M), dominated by counterparty credit spread risk.
- Greeks are complete for the book and for the exposure measures (Section 9): PFE99 at its peak date falls by about \$3,908k per +1% on all equities and rises by about \$638k per +1bp on USD rates.
- We recommend confirming with Capitolis whether any trade is margined, and on what terms; that fact matters more than any further modelling refinement.
- Use Latin Hypercube sampling with $N = 5,000$ for reporting, $N = 1,000$ for iteration and $N = 10,000$ for limit sign-off (Section 5.5).
- Next engineering steps: wire the JPY Hull-White factor into JPY equity/FX drift; a two-factor or regime-aware rate model if a long JPY curve history becomes available; a vectorised pricer reimplementations if scenario counts must exceed $\sim 10,000$; a model-risk study varying α , vols and correlation.

A Glossary

Term	Meaning
Counterparty credit risk (CCR)	Loss risk if the other side of a derivative defaults while the contract is in our favour
MTM / NPV	Mark-to-market / net present value: what the trade is worth today
Netting set	Trades under one legal netting agreement; combined before taking $\max(V,0)$
EE, PFE, MPE	Expected exposure (mean), potential future exposure (percentile), maximum PFE over time
Median PFE	50th percentile of exposure at a date
Total return swap (TRS)	Swap exchanging an asset's total return for a funding rate
Compo	Payoff in one currency on an asset quoted in another, so both asset and FX move the value
SOFR / OIS	US overnight risk-free rate / overnight index swap; the discount curve
TONA	Tokyo Overnight Average rate, JPY's overnight risk-free rate
Hull-White	Gaussian mean-reverting short-rate model that fits today's curve exactly
Mean reversion (a)	Speed at which a rate is pulled back toward its long-run path
GBM	Geometric Brownian motion: lognormal price dynamics
Cholesky	Factorisation $C = L L'$ used to create correlated random draws
PCA / factor model	Explain most co-movement with a few principal factors plus idiosyncratic noise
Latin Hypercube / Sobol	Stratified / low-discrepancy sampling schemes that reduce Monte Carlo noise
Common random numbers	Reuse identical random draws across compared runs so differences are not noise
MPOR	Margin period of risk: delay between last margin call and close-out (typically 10 business days)
CSA / VM / threshold	Credit support annex / variation margin / uncollateralized amount before margin is called
Delta, gamma	First and second sensitivity of a value or exposure measure to a market factor (here per +1% spot, per +1bp rate)
DV01	Change in value for a +1bp move of the interest rate curve, parallel or by tenor bucket
Vega	Sensitivity to volatility (here per +1% relative shift)
CVA	Credit valuation adjustment: the market price of counterparty default risk on a derivatives portfolio
SA-CVA	Basel standardised approach for CVA capital: risk-weighted CVA sensitivities aggregated by bucket and risk class (MAR50)
Wrong-way risk	Positive dependence between exposure and the counterparty default probability (not modelled here)
SIMM	ISDA Standard Initial Margin Model: sensitivity-based initial margin for uncleared derivatives, calibrated to a 99% 10-day loss (not implemented here)
Pillar dates	Standard curve tenor points: O/N, T/N, 1W, 2W, 1M ... 10Y
Risk-neutral measure	Pricing measure in which discounted assets are martingales; drift = $r - q$

B Data sources and lineage

Every external input, where it came from, how it was obtained and what it is used for. Raw pulls are stored under data/raw/ (excluded from version control); licensed data is never redistributed.

Input	Source	How obtained	Used for	Status
Trades (16) and underlyings	Capitolis: trade_data/*.csv	Supplied files	Trade definitions, baskets, bonds	Given
Pricing library	Capitolis: capitolis_pricers	Supplied package (standard library only)	All trade valuation (npv), curves, day counts	Given; independently reviewed
USD SOFR futures (SR3, SR1)	CME Globex via Databento (dataset GLBX.MDP3)	Paid API; key only in an environment variable	USD discount curve (bootstrapped by us), Hull-White fit, futures-vol mean-reversion cross-check	Live, validated against Treasury.gov

Input	Source	How obtained	Used for	Status
Treasury par yield curve	U.S. Treasury (treasury.gov)	Public download	Independent check of the SOFR curve	Validation only
Equity spots, dividends, 3y price history (37 names)	yfinance (Yahoo Finance)	Public API, keyed by ISIN	GBM start values and drift, realised volatility, correlation	Live
USDJPY spot and 3y history	yfinance	Public API	FX start value, volatility, correlation	Live
SOFR level history	FRED (series SOFR)	Public CSV	Rate volatility; USD-JPY rate correlation	From April 2018
TONA (JPY overnight rate), 1998 to date	Bank of Japan Time-Series Data Search API (DB FM01, series STRDCLUCON)	Public API	JPY rate volatility; USD-JPY correlation; negative-rate evidence	Public
JGB par yields 1Y-40Y, 1974 to date	Japan Ministry of Finance	Public CSV (browser-like request headers)	Test of JPY mean reversion	Public
JPY OIS par-rate history (MUTKCALM overnight plus 35 OIS tenors, 5 Oct 2011 to 18 Sep 2026)	Bloomberg (tickers JYSO*, MUTKCALM Index)	Provided by the project team, saved as data/raw/sources/JPY_ois_history.csv	JPY zero curve (bootstrapped), JPY volatility, JPY-USD differential, mean-reversion test	Licensed: derived numbers only in the report
USD and JPY swaption cubes; USDJPY forward points	Bloomberg one-time export, snapshot 2026-08-31	Provided by the project team	USD mean reversion; JPY cross-check; FX forwards	Licensed: derived numbers only in the report
Credit spreads by rating and maturity shape	FRED: ICE BofA US option-adjusted spread indices	Public CSV	Counterparty credit spread proxy for CVA	Public
SA-CVA parameters and formulas	BIS, Basel Framework MAR50 (July 2020 revisions)	Public PDF	Risk weights, correlations, aggregation	Public
Equity sector, country, market cap	yfinance	Public API	SA-CVA equity buckets	Live

Assumptions that are not sourced from data: counterparty ratings (BBB), LGD (60%), MPOR (10 business days), the uncollateralized status of the book, the correlation structure being static, and the JPY mean reversion being set at its lower bound ($a = 0.001$) because every calibration route gives a negative value.

C Methodology register

Component	Method	Key parameters	Where (code)
USD curve	Bootstrap of SOFR futures: implied forward rate to chained discount factors, ACT/360	33 live contracts, about 6.3y coverage, flat extrapolation beyond	market/sofr.py
USD rates	One-factor Hull-White, shifted Ornstein-Uhlenbeck, exact transition, analytic bond price	sigma 0.63% (realised SOFR); a 0.0167 (swaption cube)	models/rates.py
Mean reversion	Regression of $\ln(\text{vol})$ on tenor (vol decay); swaption cube preferred, futures and JGB as checks	USD 0.0167; futures 0.0458; JPY: all routes negative, lower bound 0.001 used	models/hw_calibration.py
JPY factor	Second Hull-White factor; curve bootstrapped from the JPY OIS par history; overnight volatility from the same file	sigma 0.267%; a = 0.001; USD-JPY rate correlation -0.04 (not significant)	models/calibration.py, market/jpy_ois.py
Equities and FX	Correlated geometric Brownian motion, exact log step, drift r-q	3y realised vols; JPY names via r_USD minus differential (about 2.6%)	models/equity_fx.py
Correlation	Static 39x39 matrix (613 aligned days), Cholesky; optional PCA factor model	5 factors explain about 46% of variance	models/equity_factor_model.py, simulation/engine.py

Component	Method	Key parameters	Where (code)
Random numbers	Latin Hypercube sampling	N = 5,000 recommended; 3,000 used for the figures	simulation/random_numbers.py
Dates	Market pillar dates plus every trade event date	42 nodes for this book	simulation/engine.py
Repricing	Full repricing of every trade in every scenario and node, 8 worker processes	About 90 ms per scenario	simulation/parallel.py
Exposure measures	Netting by counterparty, max(V,0); EE, median PFE, PFE99, MPE	PFE at the 99th percentile	exposure/aggregate.py
Collateral (hypothetical)	MPOR look-ahead on the same paths, full variation margin	10 business days, US federal holidays	exposure/collateral.py
CVA	Regulatory CVA, MAR50.32; pathwise discounting; credit-triangle PD from spreads	LGD 60%; BBB proxy; unilateral; independent	exposure/cva.py, models/credit.py
SA-CVA	CVA bump sensitivities with common random numbers, MAR50 aggregation	1,000 scenarios; 21 simulations; m_CVA = 1	exposure/sa_cva.py, scripts/run_sa_cva.py
Greeks	Bump-and-reprice with common random numbers; equity and FX bumps reuse paths (exact GBM rescaling, subset repricing); rate and vol bumps re-simulate	+1% spot, +1bp rates (parallel and 8 buckets), +1% vol; N = 2,000	greeks/book.py, greeks/exposure.py, scripts/run_greeks.py
Precision	Bootstrap resampling of a large pool; 1/sqrt(N) extrapolation	Relative SE of PFE99 about 1.0% at N = 5,000 at the worst date	scripts/convergence_study_tail.py
Validation	t=0 self-check, martingale test, delta-normal VaR, stress test, unit tests	82 automated tests	scripts/, tests/

D Code map

Package	Contents
market/	sofr, equities, fx, vols, correlations (data pulls and inputs); boj, mof_jgb, bloomberg (JPY data); credit_spreads, equity_buckets (CVA inputs)
models/	rates (Hull-White), equity_fx (GBM), calibration and hw_calibration (parameters), equity_factor_model (PCA), credit (PD from spreads)
simulation/	engine (grid, paths, MPOR nodes), random_numbers (five sampling schemes), parallel (multiprocessing repricing)
greeks/	book (t=0 Greeks by trade and netting set), exposure (sensitivities of EE, median PFE, PFE99), bumps (curve and parameter bump helpers)
exposure/	aggregate (netting, EE, median PFE, PFE, MPE), collateral (MPOR-shifted), cva (regulatory CVA), sa_cva (Basel aggregation)
scripts/	run_simulation, run_mpor_comparison, run_sa_cva, generate_report_data, build_report (LaTeX), build_exec_deck, and the convergence, variance-reduction, VaR, martingale and stress benchmarks
tests/	82 tests: engine, time grid, MPOR, negative rates, factor model, BOJ and MOF data, tail convergence, median PFE, CVA and SA-CVA

E Reproducing the results

```
python scripts/generate_report_data.py --scenarios 3000 # ~15 min: simulation, arrays for figures
python scripts/build_report.py # this PDF
python scripts/run_simulation.py --scenarios 3000 # printed EE/PFE99/MPE profiles
python scripts/run_mpor_comparison.py # hypothetical collateral demo
python scripts/run_sa_cva.py --scenarios 1000 # CVA and SA-CVA capital (21 runs)
python scripts/run_greeks.py --scenarios 2000 # book and exposure Greeks (about 75 min)
python -m pytest tests # 82 tests
```

Key modules: src/risk_engine/models (rates, equity_fx, calibration, hw_calibration, equity_factor_model), simulation (engine, random_numbers, parallel), exposure (aggregate, collateral), market (sofr, boj, mof_jgb, bloomberg, vols, correlations).